# Data Collection and Analysis: PGR215

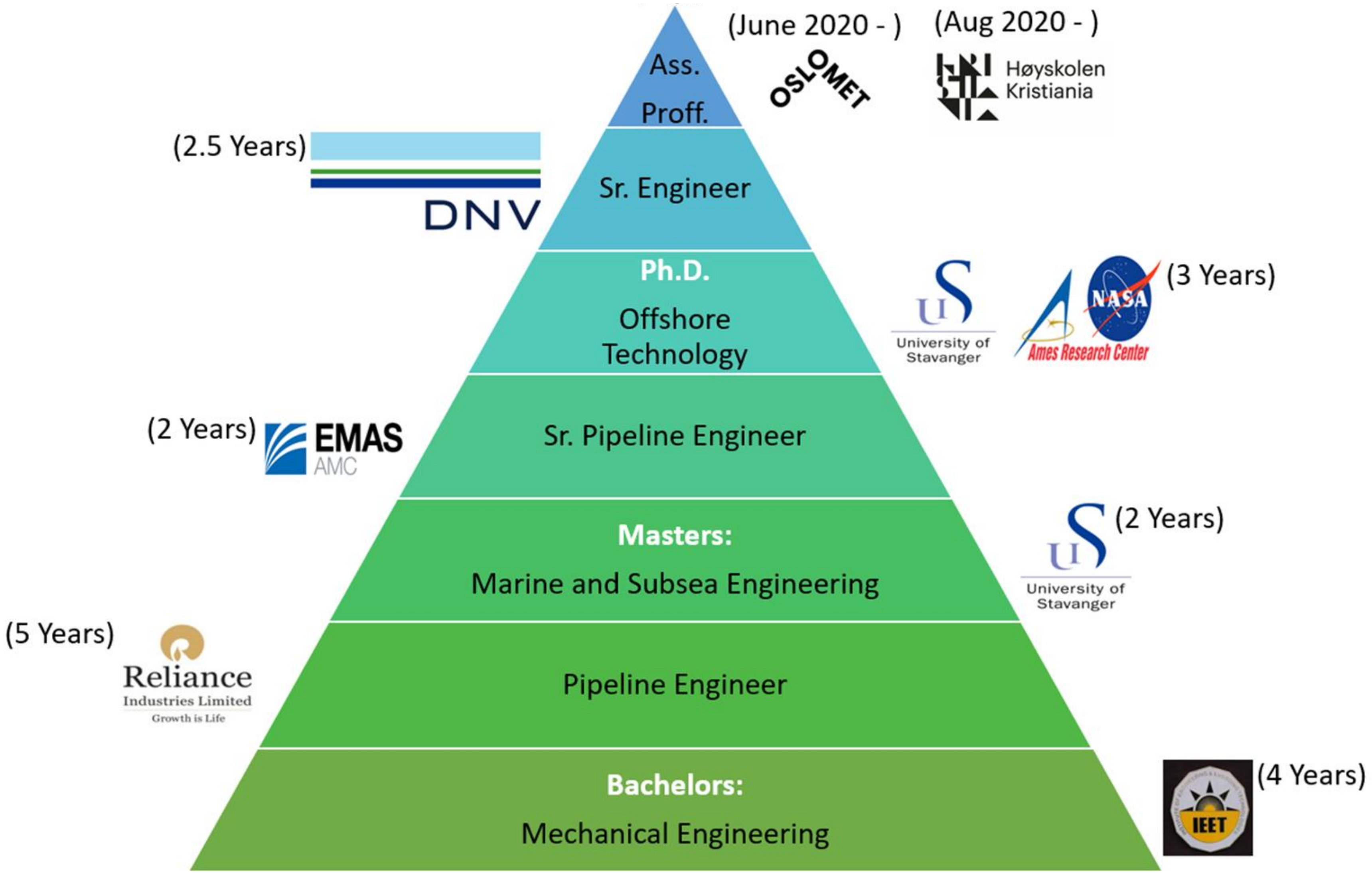
## Introduction to Modern Data Ecosytem

Dr. Arvind Keprate

# Topics for Today

- What is Data?
- What is Big Data?
- Vs of Big Data
- Big Data Life Cycle
- Learn about Databases and Different Large File Formats
- Setting Up Virtual Machine
- Test Your Knowledge

# Professional Introduction



# Important Info

- 4to 6 weeks individual portfolio exam.
- Grade Scale A to F.
- Prerequisites: Python and SQL

## Learning Outcomes

### Knowledge

The student...

- understands how to effectively collect data and clean it
- understands and assess how data influences algorithms, and being able to counter these influences with different methods or changes in data collection.
- can summarize the differences between CPU and GPU computing
- demonstrates a deep understanding of GPU computing in the context of AI

### Skills

The student...

- can select and apply an efficient method for data collection
- can select and apply efficient methods for data preparation and cleaning
- can compare the performance of running programs on CPUs versus GPUs and profile performance

### General competence

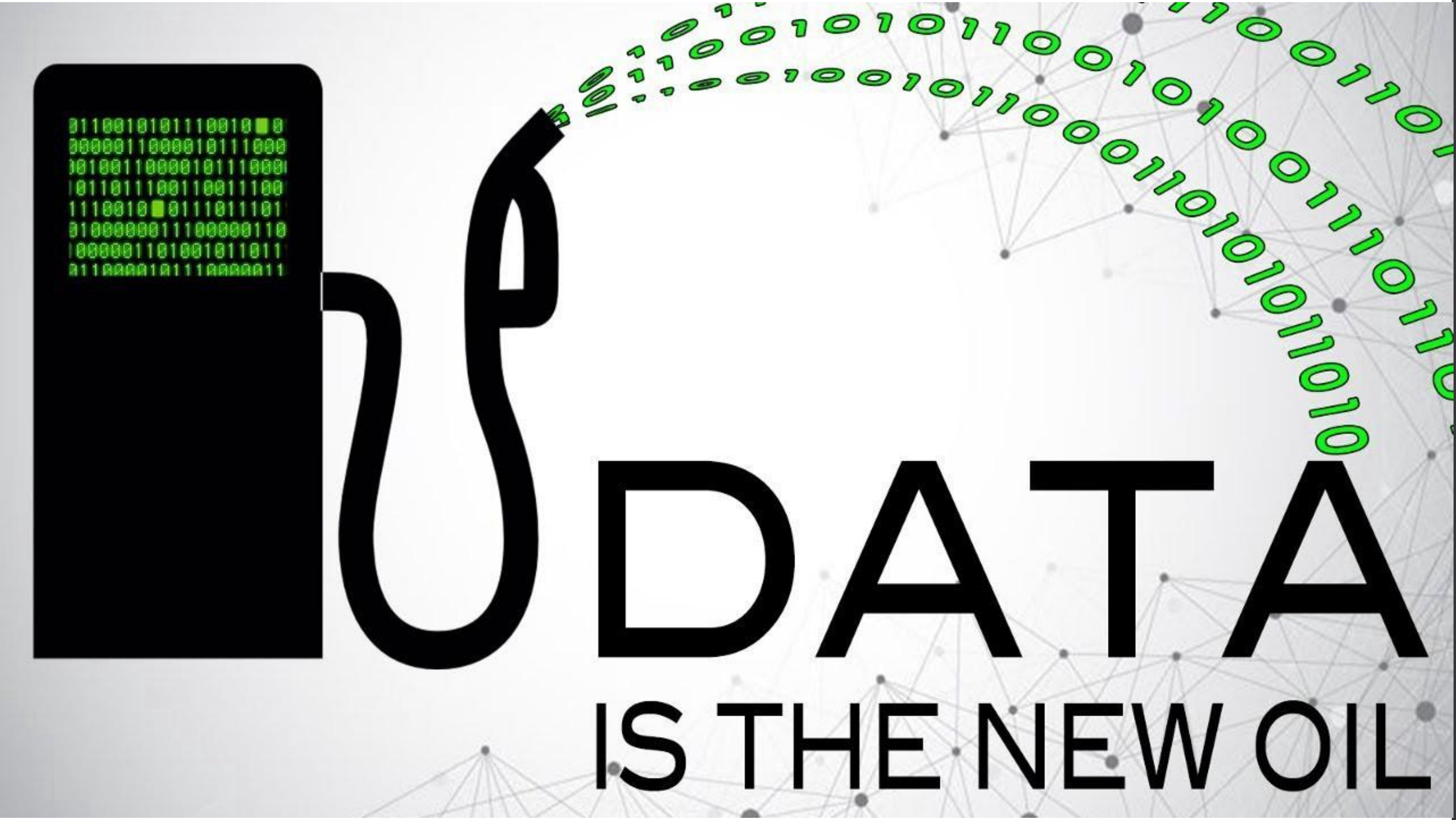
The student...

- can discuss theoretical and practical aspects of data collection
- can discuss methods and theoretical approaches for data cleaning
- can discuss theoretical aspects of parallel computing and GPU architectures

# Important Concepts for Course

1. Understanding Modern Data Ecosystem and Emerging Technologies
    - Big Data
    - Cloud Computing
    - Machine Learning
2. Data Extraction Using:
    - Web Scraping
    - API
3. Data Storage Using:
    - SQL Databases
    - NoSQL Databases

3. Data Analysis Using:
    - SQL
    - Python: Pandas, Spark (Big Data)

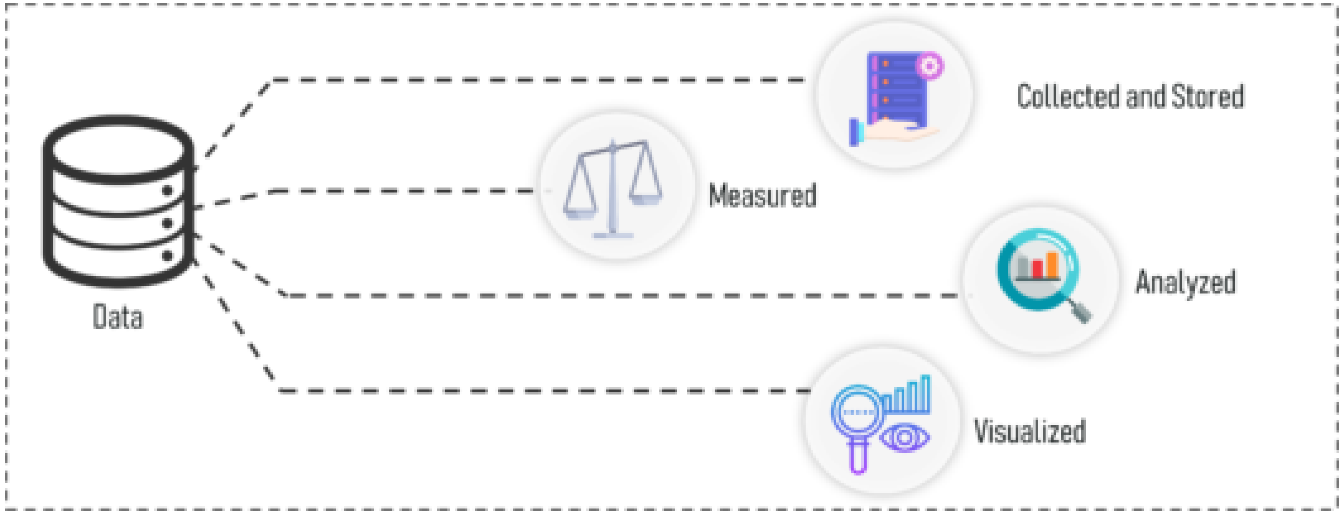

DATA
IS THE NEW OIL

# What is Data?

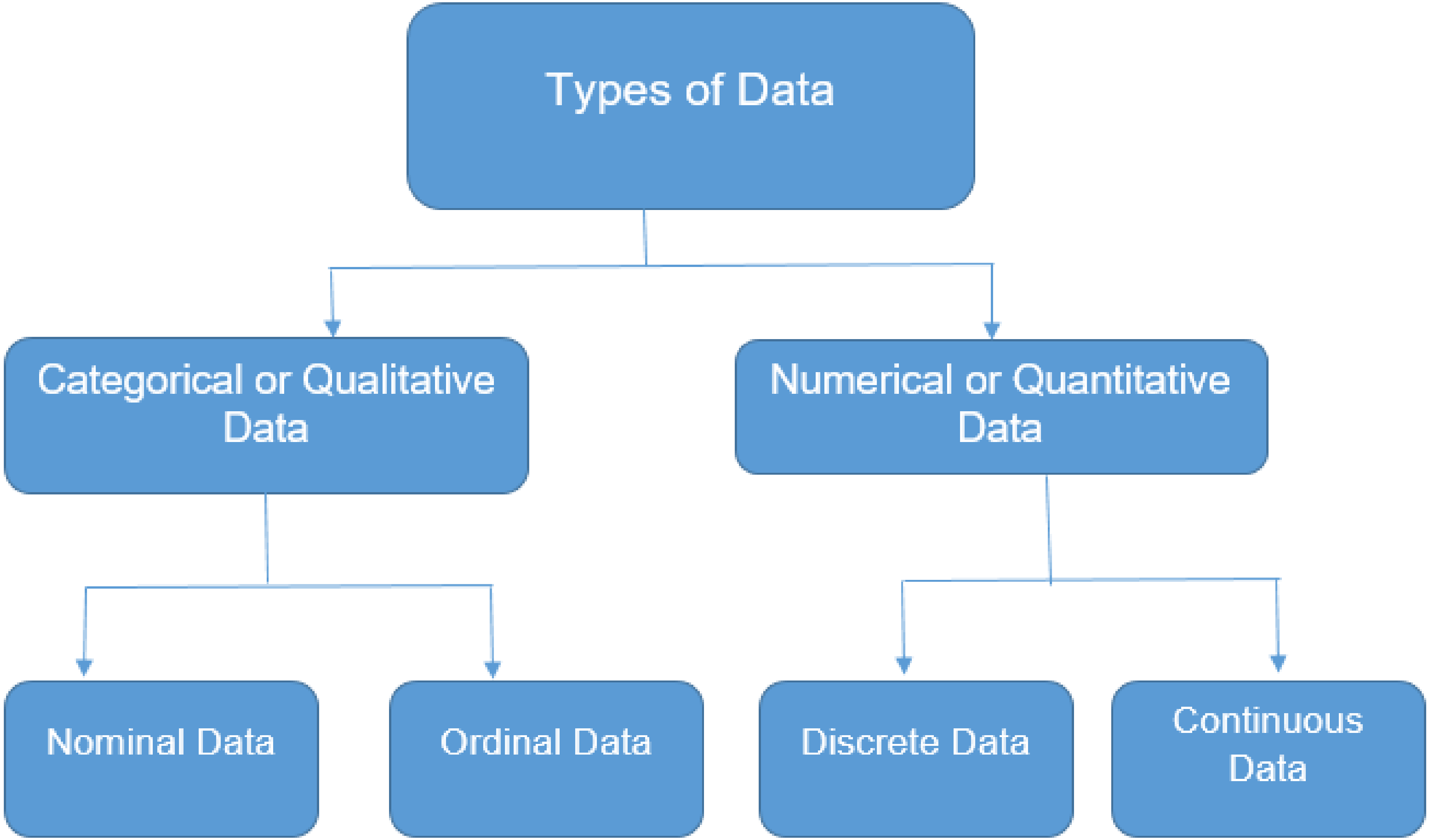
- Definition:
  - Originally, data is plural for "datum", a Latin word
  - a "datum" is a single factual, a single entity, a single point of matter.
  - Datums are most often called "**data points**".
  - *Data* represents a collection of *data points*.
    - We speak also of **datasets** instead of data (so a dataset is a collection of data points).
  - Today, "data" is used in a singular or plural form.

    -> "My data is…", but we sometimes still hear "My data are…"

- Data can be defined as things known or assumed as facts, making the basis of reasoning or calculation.

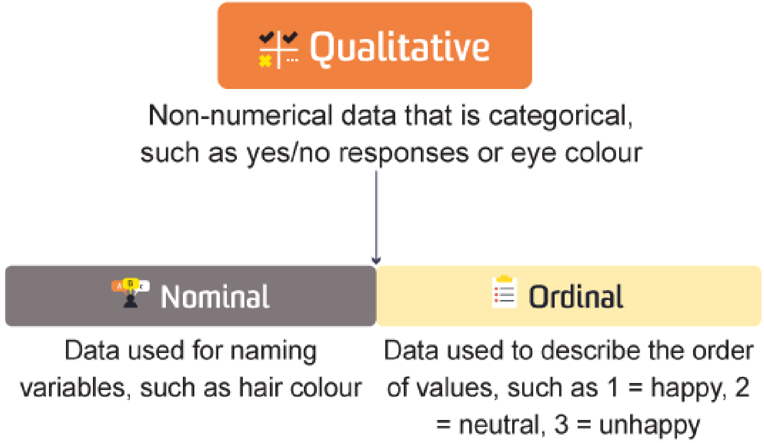


# Types of Data



# Types of Data

**Qualitative Data:** *Qualitative data deals with characteristics and descriptors that can't be easily measured, but can be observed subjectively. Qualitative data is further divided into two types of data:*




- *Nominal Data:* Data with no inherent order or ranking such as gender or race.


| Gender |
|---|
| Male |
| Female |
| Male |
| Male |

- *Ordinal Data:* Data with an ordered series of information is called ordinal data.


| Customer ID | Rating |
|---|---|
| 001 | Good |
| 002 | Average |
| 003 | Average |
| 004 | Bad |

https://aprendeconalf.es/teaching/statistics/manual/introduction/

# Types of Data

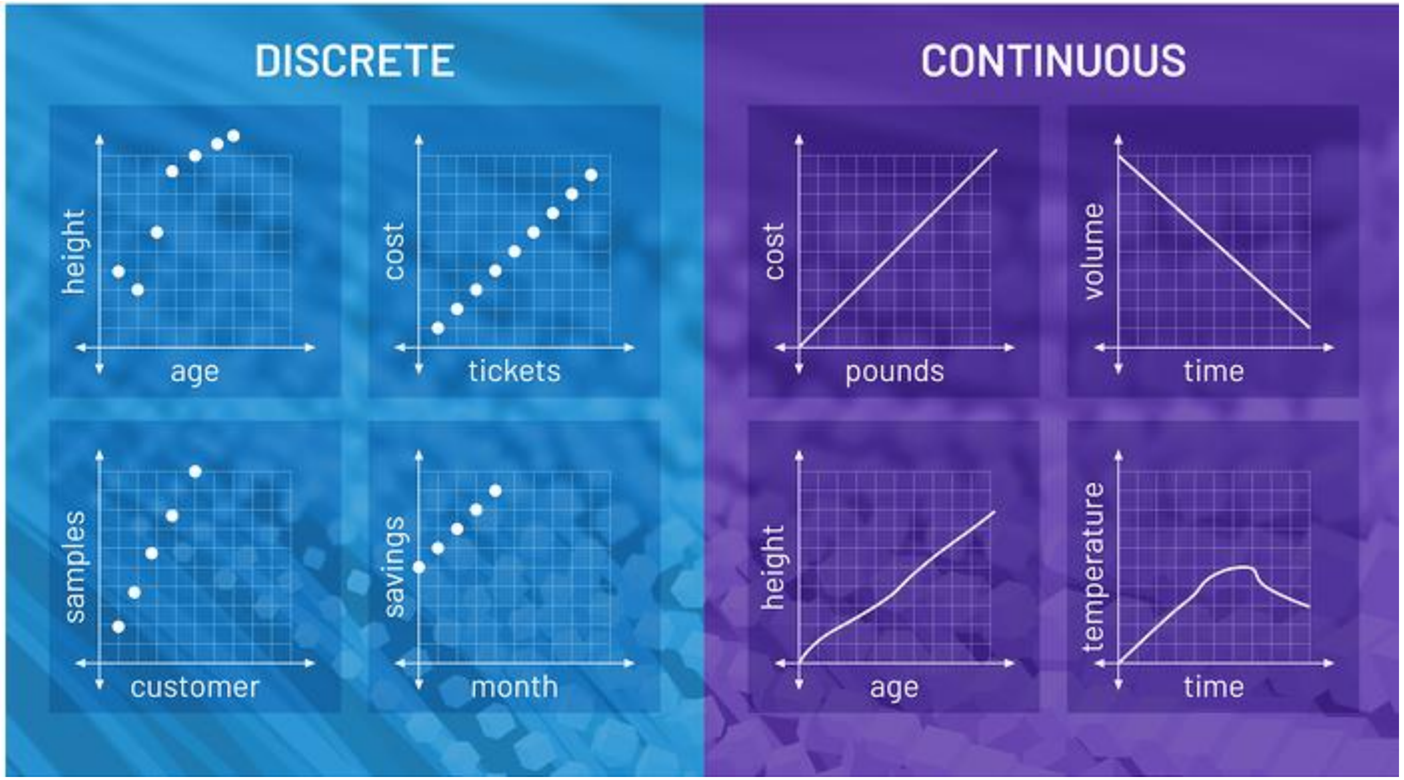
**Quantitative Data:** *Quantitative data deals with numbers and things you can measure objectively. This is further divided into two:*

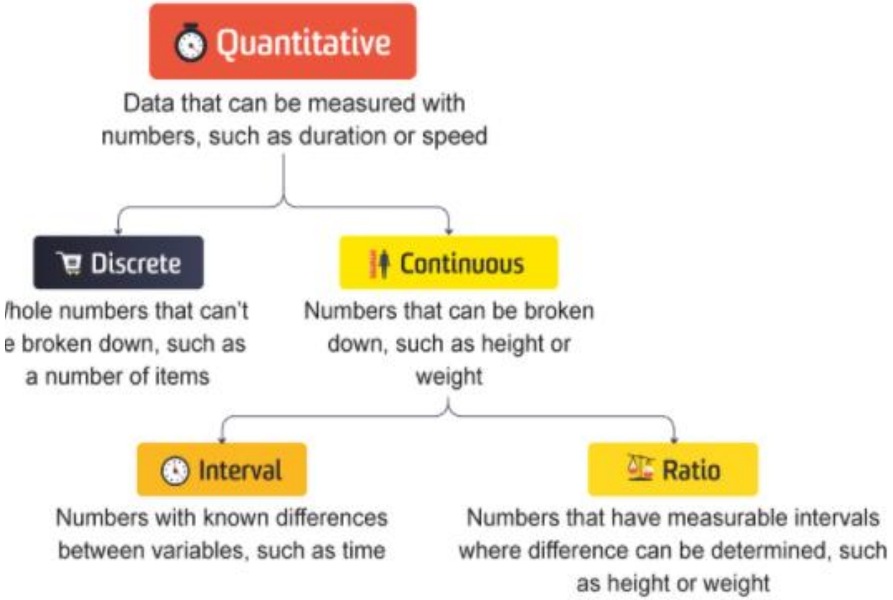
- *Discrete Data:* Also known as categorical data, it can hold a finite number of possible values.

Example: Number of students in a class.

- *Continuous Data:* Data that can hold an infinite number of possible values.

Example: Weight of a person.



# How to Choose Appropriate Type of Variable?

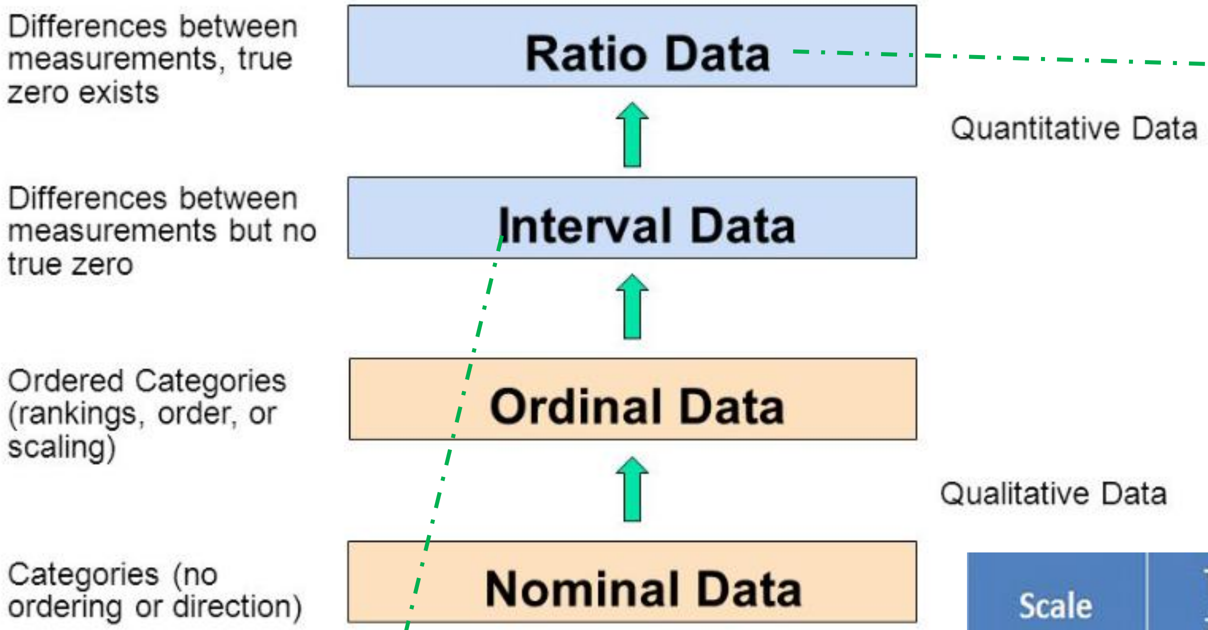
Sometimes a characteristic could be measured in variables of different types.

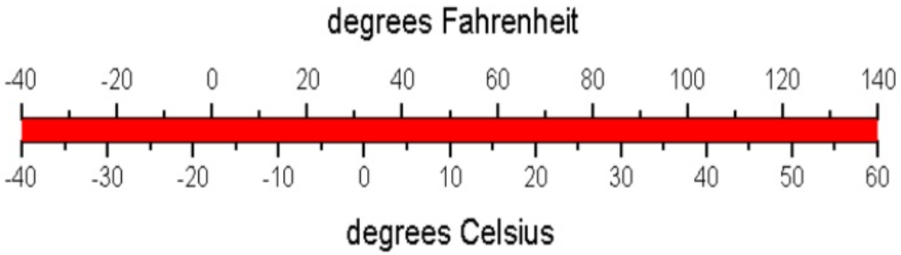
**Example.** Whether a person smokes or not could be measure in several ways:

- Smokes: yes/no. (Nominal)
- Smoking level: No smoking/unusual/moderate/quite/heavy. (Ordinal)
- Number of cigarettes per day: 0,1,2,…(Discrete)

In those cases quantitative variables are preferable to qualitative, continuous variables are preferable to discrete variables and ordinal variables are preferable to nominal, as they give more information.

# Four Scales of Measurement?




| Scale | True Zero | Equal Intervals | Order | Category | Example |
|---|---|---|---|---|---|
| Nominal | No | No | No | Yes | Marital Status, Sex, Gender, Ethnicity |
| Ordinal | No | No | Yes | Yes | Student Letter Grade, NFL Team Rankings |
| Interval | No | Yes | Yes | Yes | Temperature in Fahrenheit, SAT Scores, IQ, Year |
| Ratio | Yes | Yes | Yes | Yes | Age, Height, Weight |

Image Source: https://www.graphpad.com/support/faq/what-is-the-difference-between-ordinal-interval-and-ratio-variables-why-should-i-care/

# Trivia Time

Each scale is represented once in the list below.

- N • Favorite candy bar
- R • Weight of luggage
- I • Year of your birth
- O • Egg size (small, medium, large, extra large, jumbo)

Each scale is represented once in the list below.

- O • Military rank
- R • Number of children in a family
- N • Jersey numbers for a football team
- I • Shoe size

# Analog vs. Digital Data

Analog Data

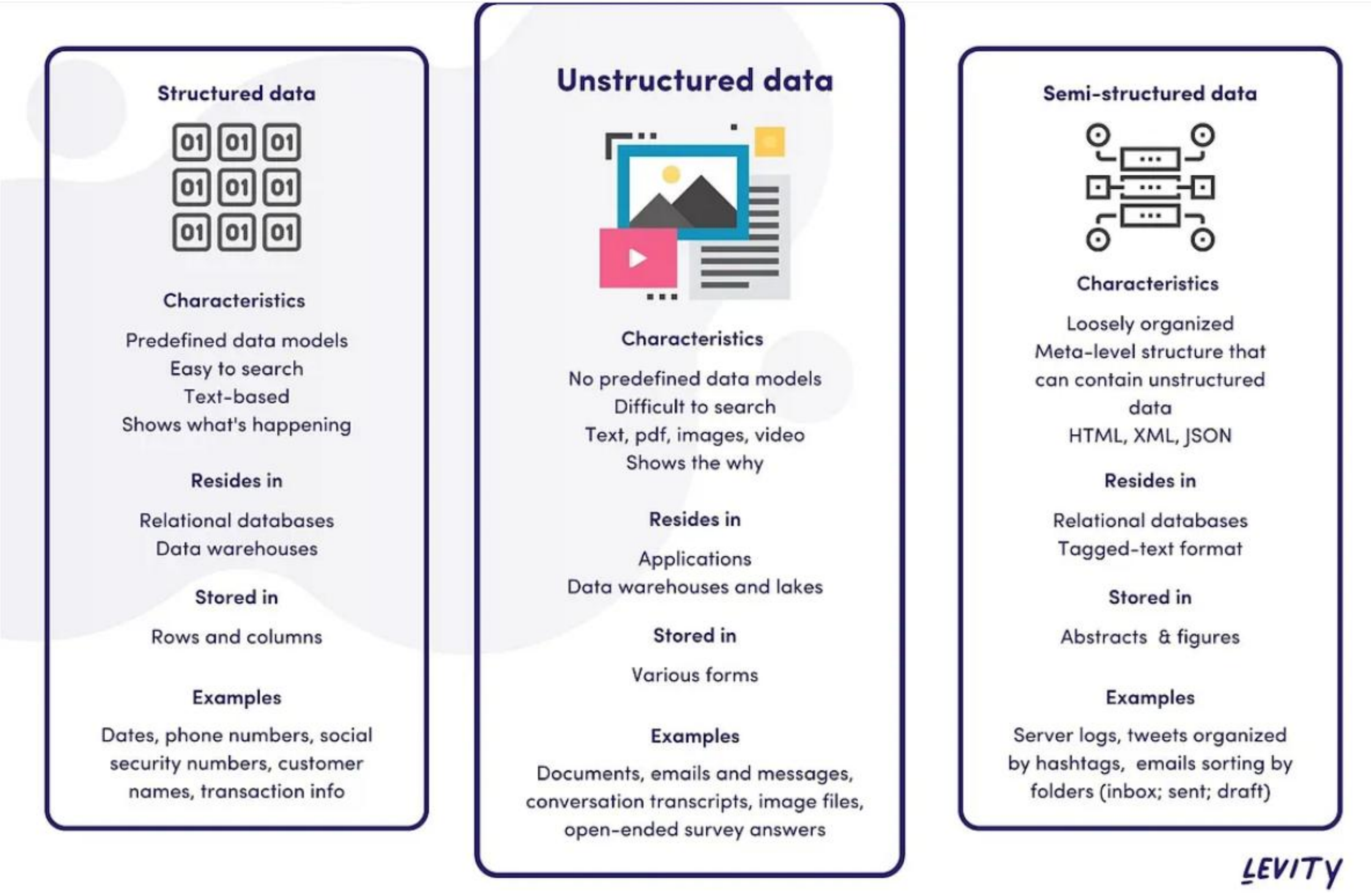
Digital Data

# Trivia Time

Which of the following are examples of digital data? Choose all that apply.

- [x] A sales receipt sent by email
- [ ] A 45 rpm record of a pop song
- [x] A customer star rating of a trip with a rideshare service
- [ ] The sight of a sunset
- [ ] Your impression of a job applicant

# Structured vs. Unstructured vs. Semi-Structured Data



# Sample Structured Data

**Column Names = Attributes**

| ID | Clump | UnifSize | UnifShape | MargAdh | SingEpiSize | BareNuc | BlandChrom | NormNucl | Mit | Class |
|---|---|---|---|---|---|---|---|---|---|---|
| 1000025 | 5 | 1 | 1 | 1 | 2 | 1 | 3 | 1 | 1 | benign |
| 1002945 | 5 | 4 | 4 | 5 | 7 | 10 | 3 | 2 | 1 | benign |
| 1015425 | 3 | 1 | 1 | 1 | 2 | 2 | 3 | 1 | 1 | malignant |
| 1016277 | 6 | 8 | 8 | 1 | 3 | 4 | 3 | 7 | 1 | benign |
| 1017023 | 4 | 1 | 1 | 3 | 2 | 1 | 3 | 1 | 1 | benign |
| 1017122 | 8 | 10 | 10 | 8 | 7 | 10 | | 7 | 1 | malignant |
| 1018099 | 1 | 1 | 1 | 1 | 2 | 10 | 3 | 1 | 1 | benign |
| 1018561 | 2 | 1 | 2 | H | 2 | 1 | 3 | 1 | 1 | benign |
| 1033078 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 5 | benign |
| 1033078 | 4 | 2 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | benign |

**Columns = Feature**

**Row = Observation**

# Sample Unstructured Data

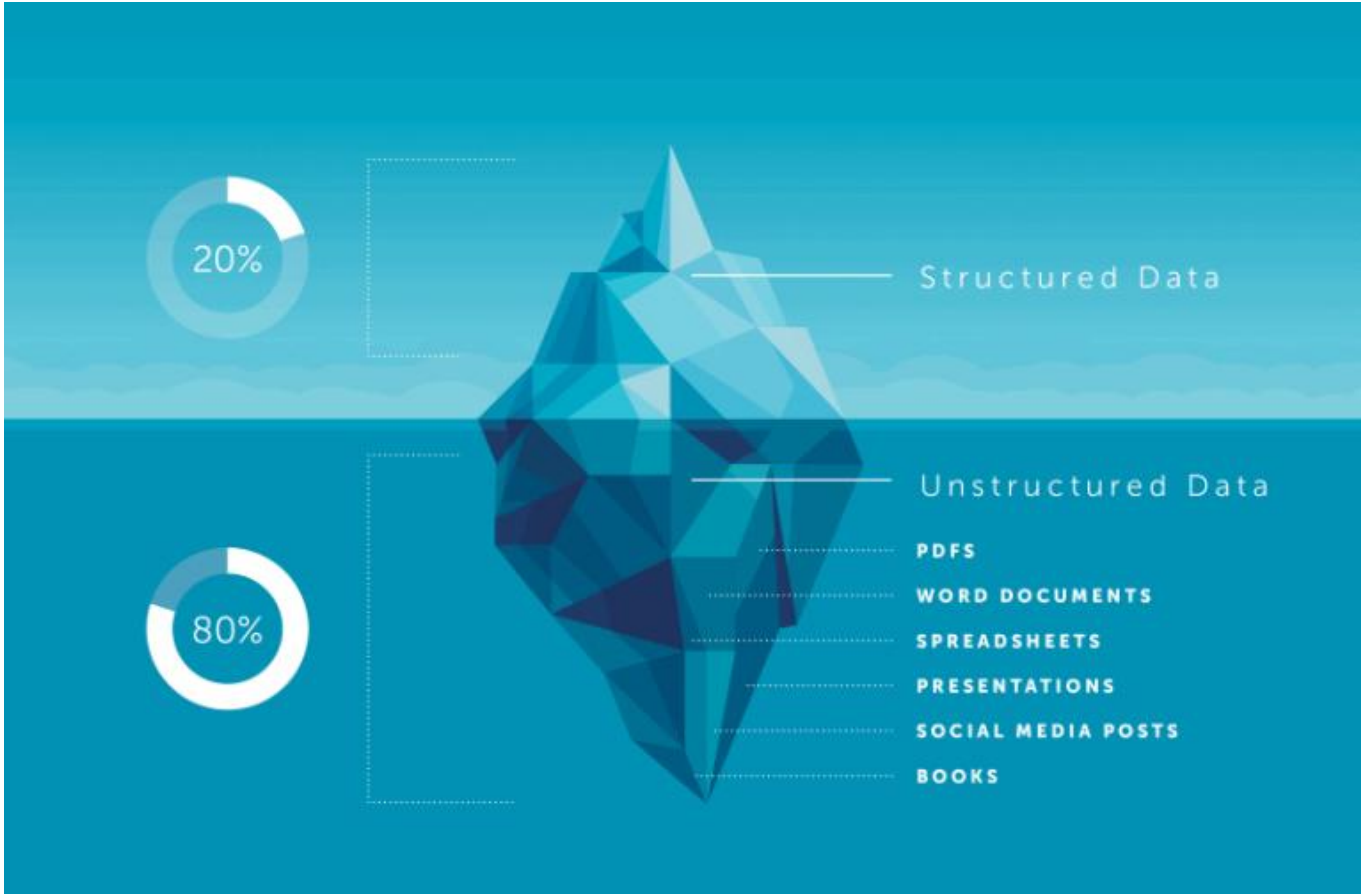


# Sample Semi-Structured Data

**JavaScript Object Notation (JSON) format**

```
## Document 1 ##
{
  "customerID": "103248",
  "name":
  {
    "first": "AAA",
    "last": "BBB"
  },
  "address":
  {
    "street": "Main Street",
    "number": "101",
    "city": "Acity",
    "state": "NY"
  },
  "ccOnFile": "yes",
  "firstOrder": "02/28/2003"
}
```

# Structured vs. Unstructured Data

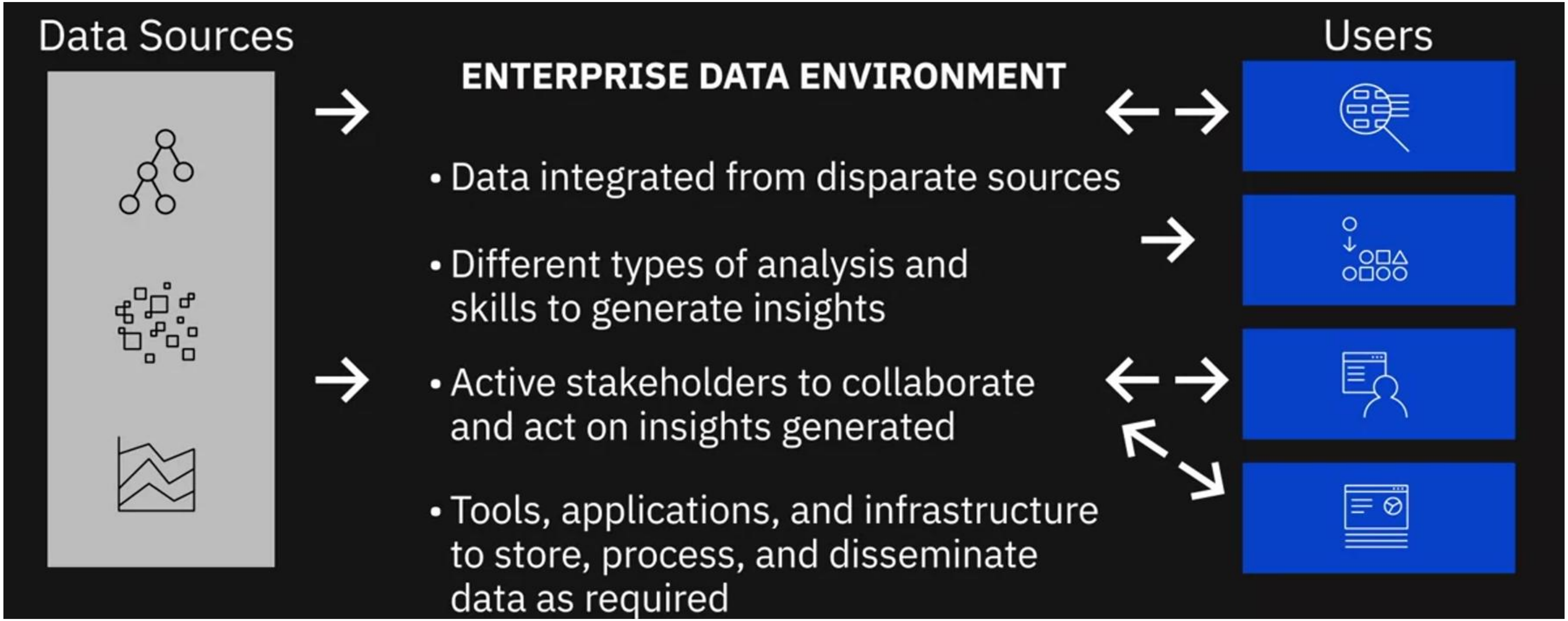



Ref: https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/

# Challenges of Unstructured Data



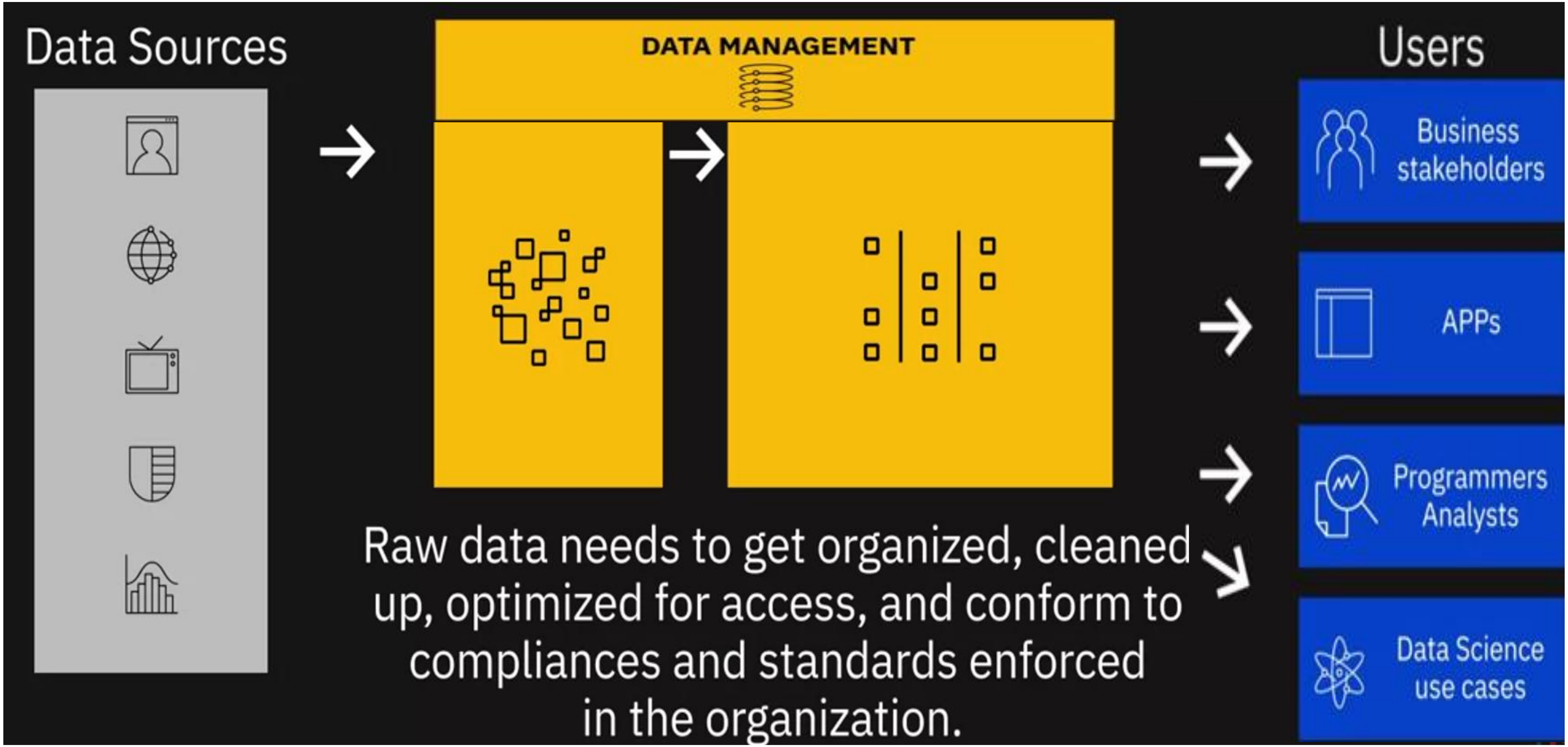
# Modern Data Ecosystem



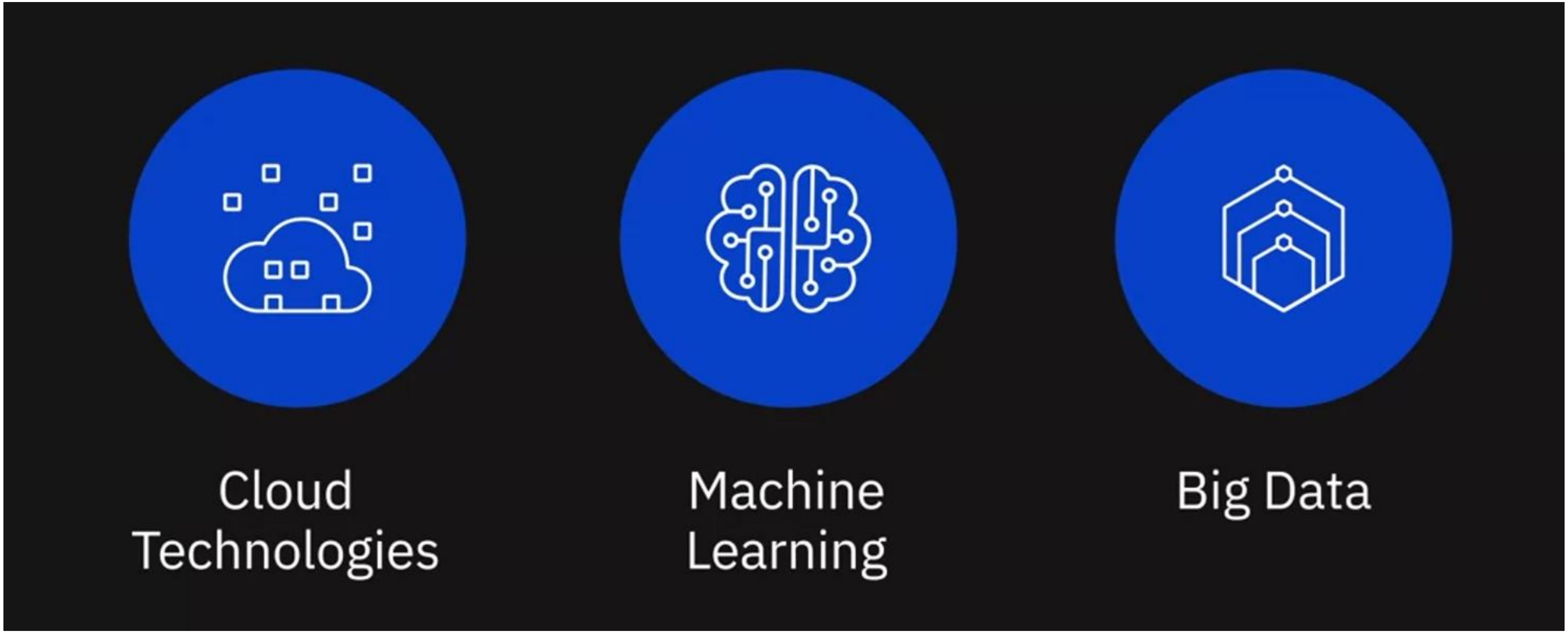

- A modern data ecosystem includes a whole network of interconnected, independent, and continually evolving entities.

- It includes data that has to be integrated from disparate sources; different types of analysis and skills to generate insights; active stakeholders to collaborate and act on insights generated; and tools, applications, and infrastructure to store, process, and disseminate data as required

# Modern Data Ecosystem



# Emerging Technologies in Data Landscape



BREAK
15MINS

# What is Big Data?


BIG
DATA
HIGH
TRENDS
INSIGHT
LOGS
TYPES
HANDLE
THINGS
BLANKET
ANY
BECOMES
DEPENDENCIES
STREAMS
TIME
ENABLE
TRIGGER
AREAS
WIRELESS
OUT
DENSITY
COLLECTION
TERM
ENHANCED
PHYSICS
HUNDREDS
TOOLS
TRANSFER
ANALYSIS
INCLUDE
CAPABILITIES
DISCOVERY
RESEARCH
SIMULATIONS
MAKING
NETWORKS
ENTERPRISE
BUSINESS
SOFTWARE
MICROPHONES
PROCESSING
TAKE
DATA
SHARING
CAPTURE
DECISION
SCIENTISTS
INTERNET
STORAGE
SPEED
PACKAGES
FINANCE
REPORT
PROCESS
BIG
CURATE
PACKAGES
GENOMICS
LARGE
CURATION
OPTIMIZATION
TERABYTES
DATA
CAPTURE
INFORMATION
SENSOR
AND
MANAGEMENT
PROCESSING
ON-HAND
PETABYTES
AMOUNT
IN
VELOCITY
VARIETY
CHALLENGES
USING
TRADITIONAL
3V
REQUIRE
FOR
RELATIONAL
METEOROLOGY
STATISTICS
DOMAIN
INFORMATICS
DIFFICULT
PROCESS
DATABASE
VOLUME
RANGE
VELOCITY
GROWS
SIMULATION
THAT
EXPLORE
GARTNER
COMPLEX
ASSETS
THAT
CAMERAS
AND
BYTES
APPLICATIONS
IS A
FORMS
ENVIRONMENTAL
TOLERABLE
CONNECTOMICS
PROCESS
PROBLEMS
BUSINESS INTELLIGENCE
NEW
SOURCES
DOUG
LANEY
MANAGE
SEARCH
FLOW
SYSTEMS
SIZE
GOVERNMENT
VISUALIZATION

"The basic idea behind the phrase 'Big Data' is that everything we do is increasingly leaving a digital trace (or data), which we can use and analyze to become smarter. The driving forces in this brave new world are access to ever-increasing volumes of data and our ever-increasing technological capability to mine that data for commercial insights."

—Bernard Marr

# What is Big Data?

# Data Sizes in Use

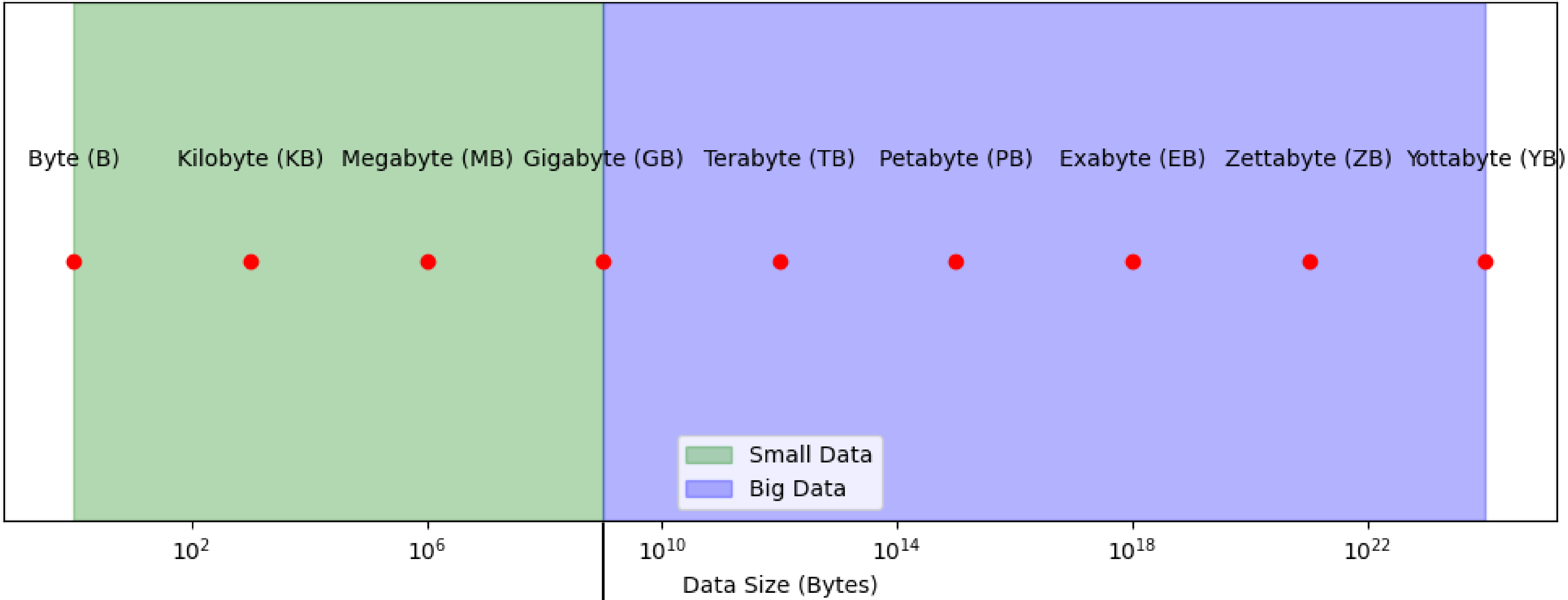
| Abbreviation | Unit | Value | Size (in bytes) |
|---|---|---|---|
| b | bit | 0 or 1 | 1/8 of a byte |
| B | bytes | 8 bits | 1 byte |
| KB | kilobytes | 1,000 bytes | 1,000 bytes |
| MB | megabyte | $1,000^2$ bytes | 1,000,000 bytes |
| GB | gigabyte | $1,000^3$ bytes | 1,000,000,000 bytes |
| TB | terabyte | $1,000^4$ bytes | 1,000,000,000,000 bytes |
| PB | petabyte | $1,000^5$ bytes | 1,000,000,000,000,000 bytes |
| EB | exabyte | $1,000^6$ bytes | 1,000,000,000,000,000,000 bytes |
| ZB | zettabyte | $1,000^7$ bytes | 1,000,000,000,000,000,000,000 bytes |
| YB | yottabyte | $1,000^8$ bytes | 1,000,000,000,000,000,000,000,000 bytes |

Ref: https://www.weforum.org/agenda/2019/04/how-much-data-is-generated-each-day-cf4bddf29f/

# Data Sizes in Use

| Data Size | Examples of Use |
|---|---|
| Kilobyte (KB) | Small text files, like a simple Notepad document or a basic email without attachments. |
| Megabyte (MB) | Photos captured by smartphones, MP3 songs, PDF documents, or small applications. For example, a 5-minute MP3 song can be around 5 MB. |
| Gigabyte (GB) | High-definition (HD) movies, larger software applications, and video games. A typical HD movie might take 1-2 GB of storage. |
| Terabyte (TB) | Storage on high-end personal computers, external hard drives, or cloud storage services. A typical external hard drive might have 1-4 TB of storage capacity. |
| Petabyte (PB) | Data centers, cloud service providers (e.g., Google, Amazon). One petabyte is enough to store around 13 years of HD video. |
| Exabyte (EB) | Global data generated by large organizations over long periods, such as internet traffic, or large-scale scientific data like from NASA satellites. |
| Zettabyte (ZB) | Total global data generated annually. For example, by 2020, it was estimated that global data storage would reach close to 44 zettabytes. |
| Yottabyte (YB) | Currently theoretical in practical usage, Yottabyte would represent the total capacity required for the largest future data infrastructures, like internet-scale data. |

# Small Data vs. Big Data




**Examples:** Personal files, small business records, data used in spreadsheets, a personal music or movie collection.

Small data refers to manageable, typically smaller datasets that can be processed with simple tools like spreadsheets or SQL databases

**Examples:** Data from social media platforms, e-commerce transactions, healthcare records, scientific research data, IoT sensor data.

Big data refers to massive datasets that are too large or complex to be processed by traditional methods and require advanced technologies like distributed storage and parallel processing.

# Small Data vs. Big Data

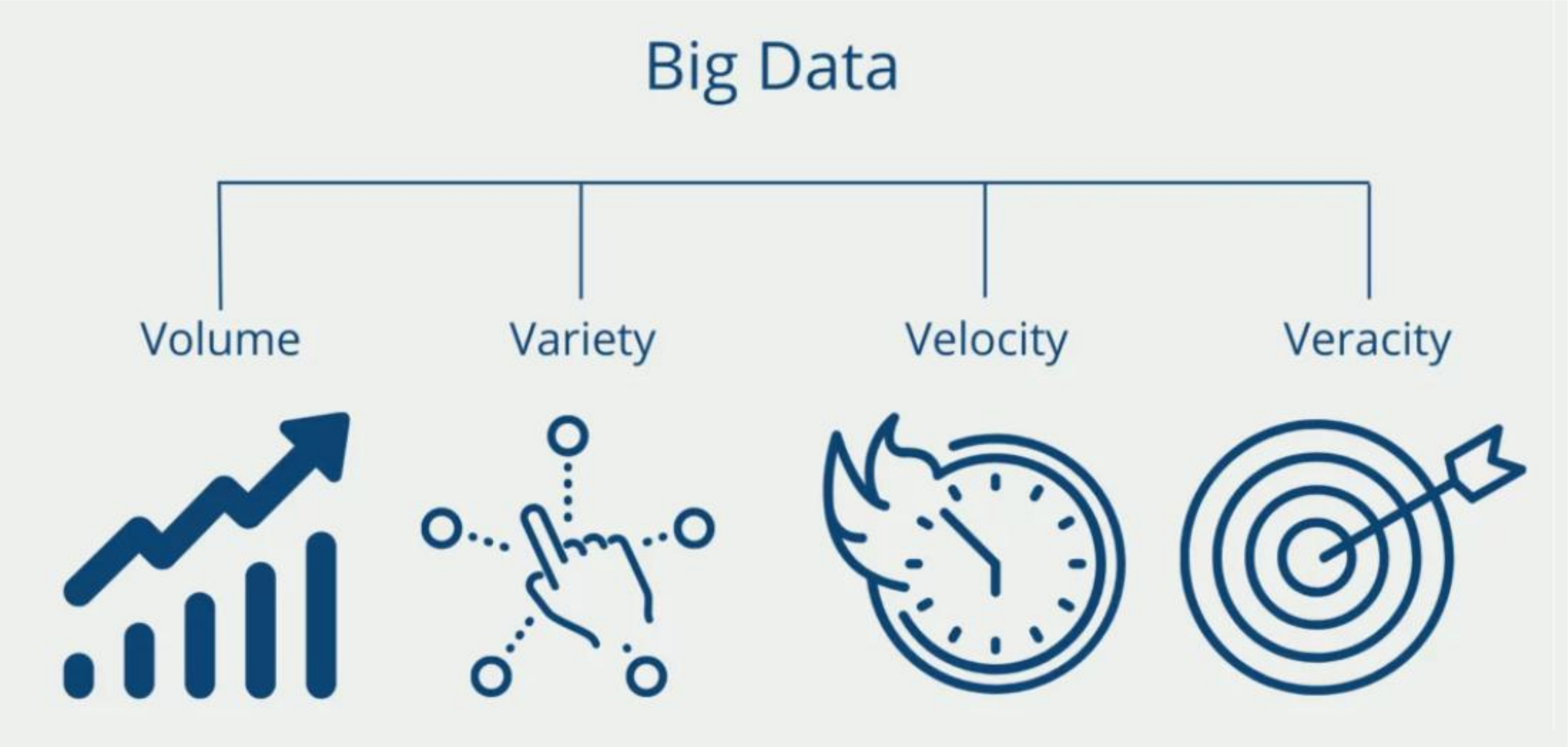
## Small Data

- Small enough for human inference
- Accumulated slowly
- Relatively consistent and structured data usually stored in known forms such as JSON and XML
- Mostly located in storage systems within Enterprises or data centers

## Big Data

- Data generated in huge volumes and could be structured, semi-structured, or unstructured
- Needs processing to generate insights for human consumption
- Arrives continuously at enormous speed from multiple sources
- Comprises any form of data including video, photos, and more
- Distributed on the cloud and server farms

# 4 Vs of Big Data




Ref: https://databasecamp.de/en/data/big-data-basics

# 4 Vs of Big Data


**Definition:** Refers to the vast amount of data generated every second across the globe. Big data is characterized by the sheer scale of the information being produced.

**Example:**

- Social media platforms like Facebook generate petabytes of data daily from posts, comments, likes, and videos.
- Google processes over 40,000 search queries every second, amounting to over 3.5 billion searches per day.
- Sensors on Internet of Things (IoT) devices generate huge datasets, such as data from smart thermostats, industrial equipment, or self-driving cars.


**Definition:** Variety refers to the different types of data being generated, from structured to unstructured and semi-structured data. Big data comes in a wide range of formats that are difficult to manage and process with traditional systems.

**Example:**

- Structured data: Customer databases in a relational format (SQL).
- Unstructured data: Social media posts, videos, images, emails, and audio files.
- Semi-structured data: Log files, XML, or JSON data from web servers that don't fit neatly into tables but have some structure.

# 4 Vs of Big Data

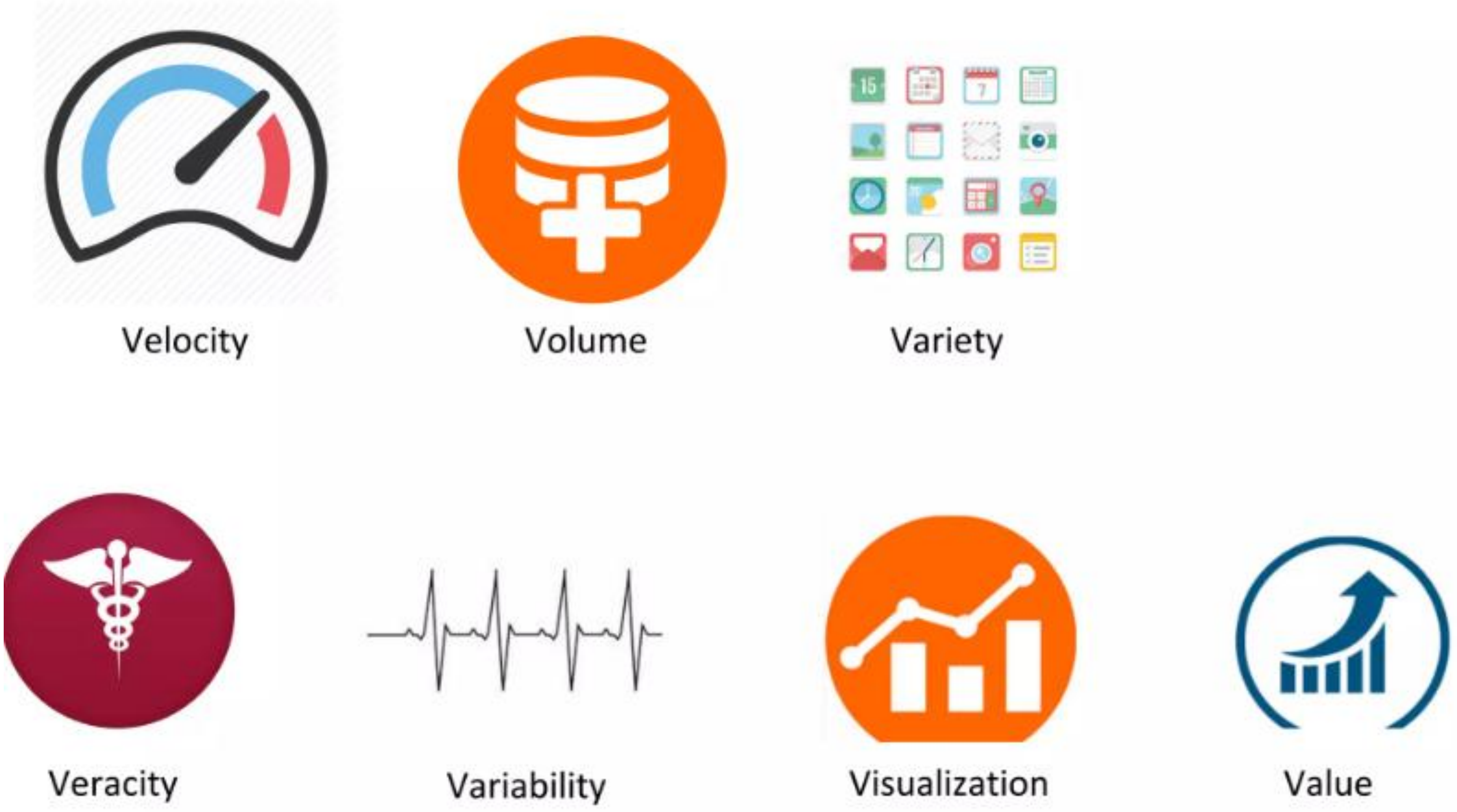

**Definition:** Velocity refers to the speed at which data is generated, collected, and analyzed. This characteristic highlights the need for real-time data processing and analysis.

**Example:**
- Twitter: Users post approximately 500 million tweets per day, requiring real-time data analysis to monitor trends, events, and sentiments.
- Stock Market Trading: High-frequency trading systems analyze large datasets in real-time to make split-second decisions on buying and selling stocks.
- Streaming services like Netflix analyze real-time user behavior (what you watch, skip, or pause) to provide personalized recommendations instantly.


**Definition:** Veracity refers to the quality, accuracy, and trustworthiness of the data. Big data often includes large amounts of unverified or noisy data, making it challenging to ensure data integrity.

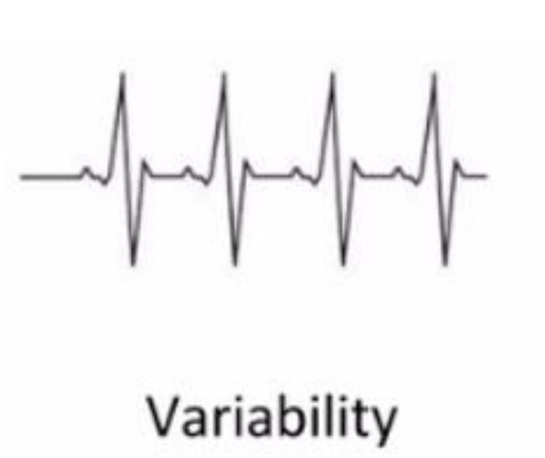
**Example:**
- Social Media Feeds: There can be a lot of noise or inaccurate information in social media feeds, like fake news, making it harder to analyze accurate trends.
- Sensor Data: IoT sensors may generate faulty or inaccurate data due to hardware issues, environmental factors, or transmission errors, which need to be filtered out during analysis.
- Medical Data: Patient records from multiple sources can contain inconsistencies or incorrect diagnoses, impacting decision-making processes in healthcare analytics.

# 7 Vs of Big Data




Ref: https://www.slideshare.net/Mahmoudyassin1/big-data-introduction-hadoop-in-details

# 7 Vs of Big Data




**Definition:** Refers to the data's changing nature and the inconsistencies in the data flow. It highlights fluctuations in data streams and unpredictable data patterns.

**Example:**

- Social Media Engagement: The number of posts, comments, or shares can fluctuate depending on trends, events, or seasons.
- E-commerce traffic: Data volume and user interactions spike during sales events like Black Friday or Cyber Monday, requiring systems to handle unpredictable data loads.
- Weather Data: IoT sensors and satellite feeds may produce erratic data due to environmental factors or equipment errors.


**Definition:** Refers to the representation of data in graphical formats, helping stakeholders understand insights derived from big data quickly and clearly.

**Example:**

- Business dashboards: Companies like Tableau or Power BI use visualization tools to represent sales data, website analytics, or marketing performance with charts, graphs, and heatmaps.
- Healthcare: Real-time patient monitoring systems visualize vital signs like heart rate, oxygen levels, and blood pressure to enable quick decisions by healthcare professionals.
- Social Networks: Platforms like LinkedIn use network graphs to represent the connections between users for understanding relationship patterns and professional connections.

# 7 V's of Big Data



Ref: https://www.collidu.com/presentation-big-data-applications

# 7 V’s of Big Data


**Definition:** Value is the benefit that organizations or individuals can extract from big data. It's the most crucial V as it emphasizes the importance of analyzing big data to extract useful insights that drive decisions, improve efficiency, or generate revenue.

**Example:**

- Retail: Companies like Amazon use customer purchasing data to optimize their product recommendations, improving sales and customer satisfaction.

- Healthcare: Analyzing big data from patient health records helps improve diagnostic accuracy, treatment planning, and even predict disease outbreaks.

- Fraud Detection: Banks and financial institutions use big data to detect patterns of fraudulent behavior and prevent theft in real-time.

- Predicitve Maintenance: Big data allows manufacturers to monitor machine health through sensor data, predicting when maintenance is required before a breakdown occurs.

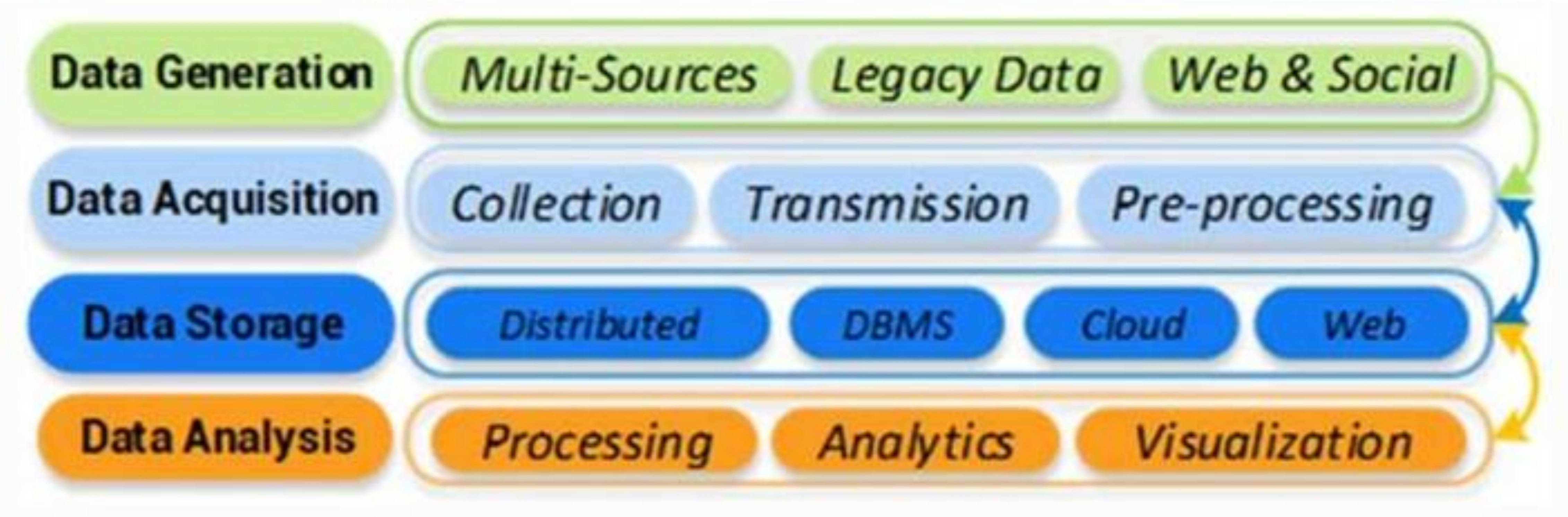
- Energy Efficiency: Big data helps track and analyze household energy consumption, helping customers and utility companies promote more efficient energy use.

- Oil and Gas Exploration: Big data aids in analyzing seismic data and geological surveys to predict the best places for oil and gas exploration.

- Urban Planning: Cities use big data for infrastructure planning, analyzing traffic patterns, public transportation usage, and demographic data to improve city layout and resource allocation.

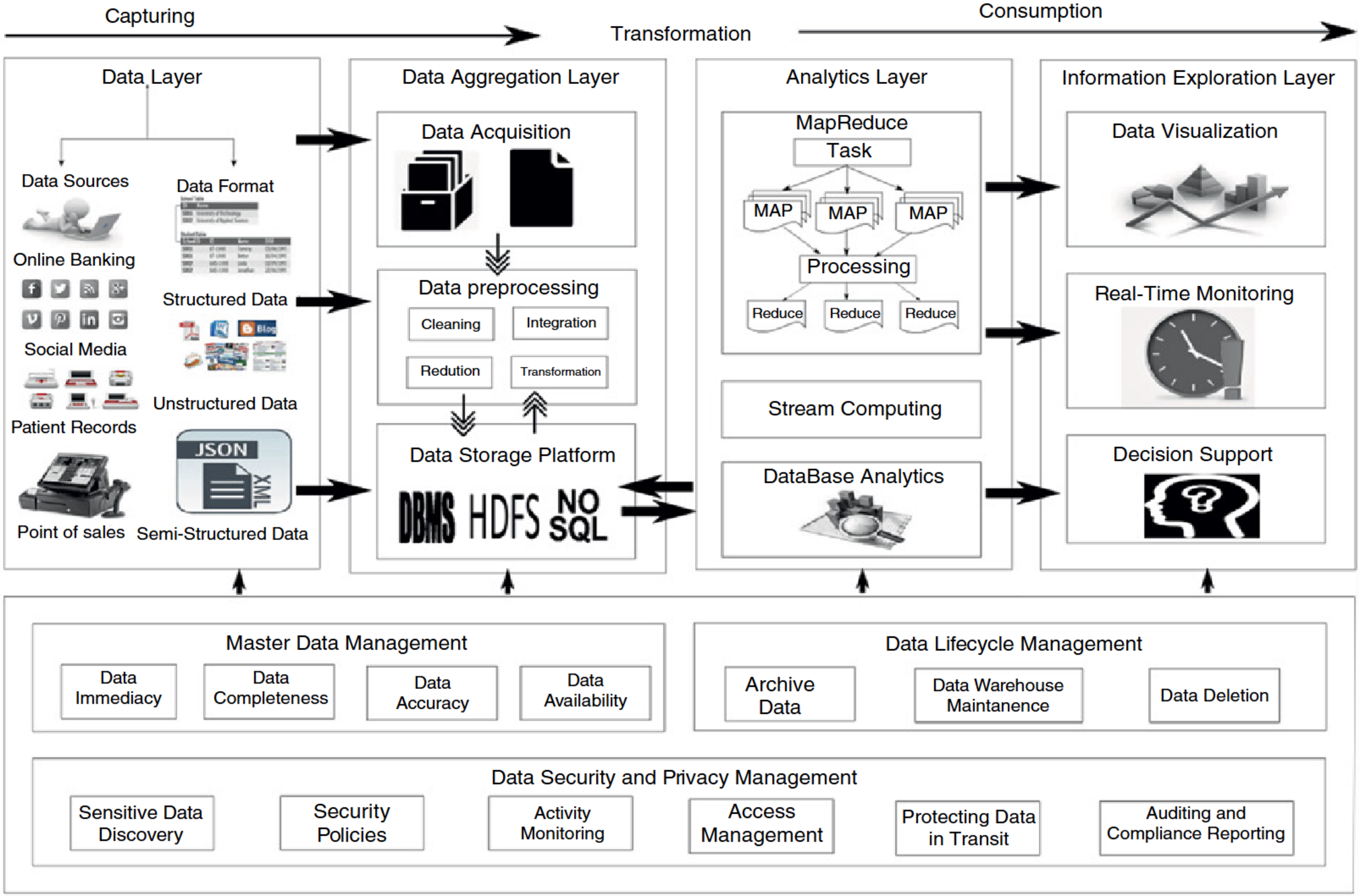
- Audience Targeting: Big data helps media companies target content to specific audiences based on demographics, location, and preferences.

# Big Data Life Cycle



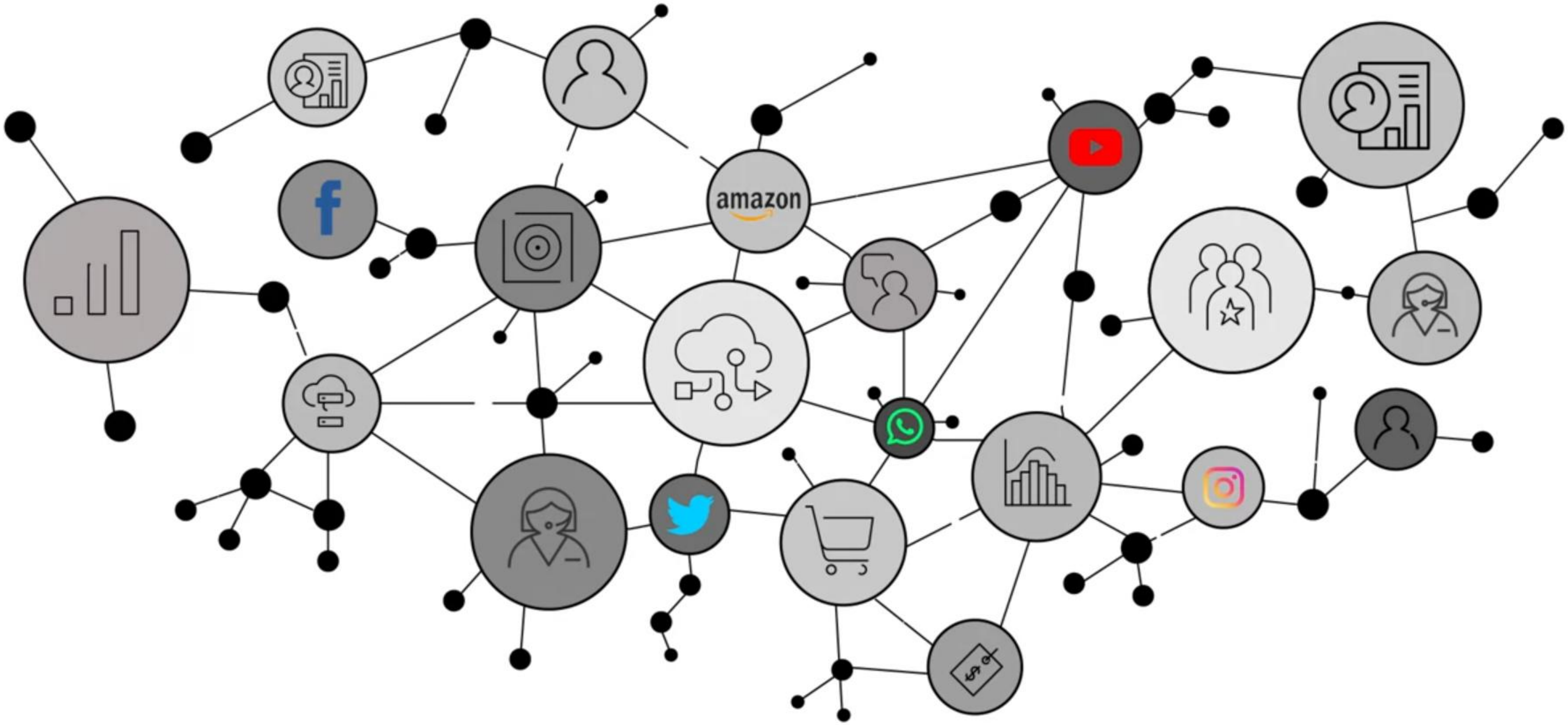

Ref: https://journalofbigdata.springeropen.com/articles/10.1186/s40537-021-00468-0

# Big Data Life Cycle




Ref: https://onlinelibrary.wiley.com/doi/book/10.1002/9781119701859

# Data Generation Sources



# Data Generation Sources



Ref: https://www.slideshare.net/Mahmoudyassin1/big-data-introduction-hadoop-in-details

# A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion - fuelled by internet of things and the use of connected devcies – are hard to comprehend, particularly when looked at in the context of one day

**500m**

tweets are sent every day

Twitter

**4PB**

of data created by Facebook, including

**350m** photos

**100m** hours of video watch time

Facebook Research

**294bn**

billion emails are sent

Radicati Group

**320bn**

emails to be sent each day by 2021

**306bn**

emails to be sent each day by 2020

**3.9bn**

people use emails

**4TB**

of data produced by a connected car

Intel

## ACCUMULATED DIGITAL UNIVERSE OF DATA


PwC

## DEMYSTIFIYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

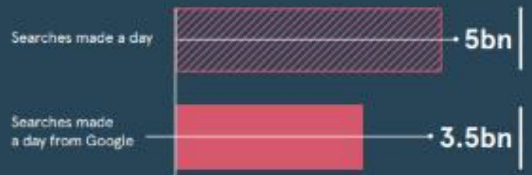
| Unit | | Value | Size |
|---|---|---|---|
| b | bit | 0 or 1 | 1/8 of a byte |
| B | byte | 8 bits | 1 byte |
| KB | kilobyte | 1,000 bytes | 1,000 bytes |
| MB | megabyte | $1{,}000^2$ bytes | 1,000,000 bytes |
| GB | gigabyte | $1{,}000^3$ bytes | 1,000,000,000 bytes |
| TB | terabyte | $1{,}000^4$ bytes | 1,000,000,000,000 bytes |
| PB | petabyte | $1{,}000^5$ bytes | 1,000,000,000,000,000 bytes |
| EB | exabyte | $1{,}000^6$ bytes | 1,000,000,000,000,000,000 bytes |
| ZB | zettabyte | $1{,}000^7$ bytes | 1,000,000,000,000,000,000,000 bytes |
| YB | yottabyte | $1{,}000^8$ bytes | 1,000,000,000,000,000,000,000,000 bytes |

*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

**65bn**

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook




Smart Insights

**463EB**

of data will be created every day by 2025

IDC

**95m**

photos and videos are shared on Instagram

Instagram Business

**28PB**

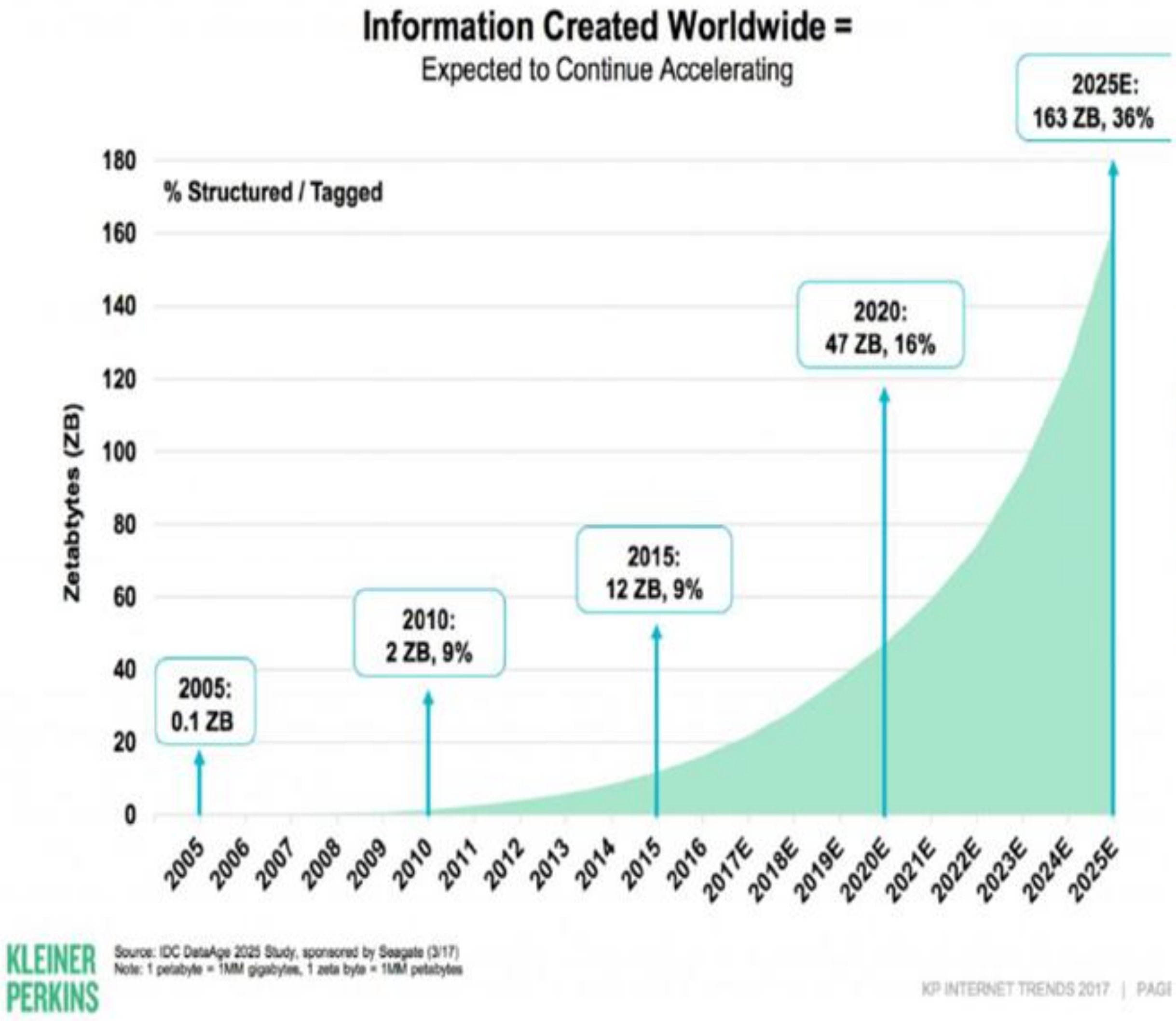
to be generated from wearable devices by 2020

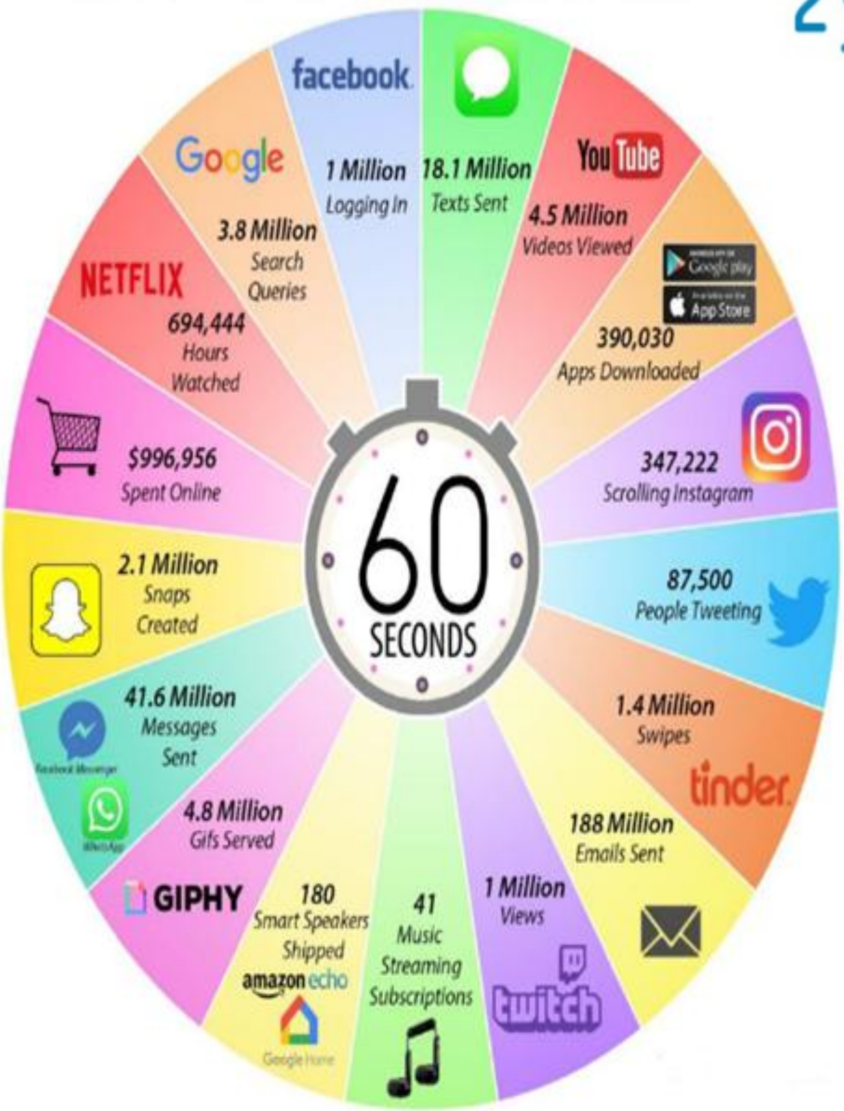
Statista

RACONTEUR

# Facts About Data

The World's data is doubling every 2 years

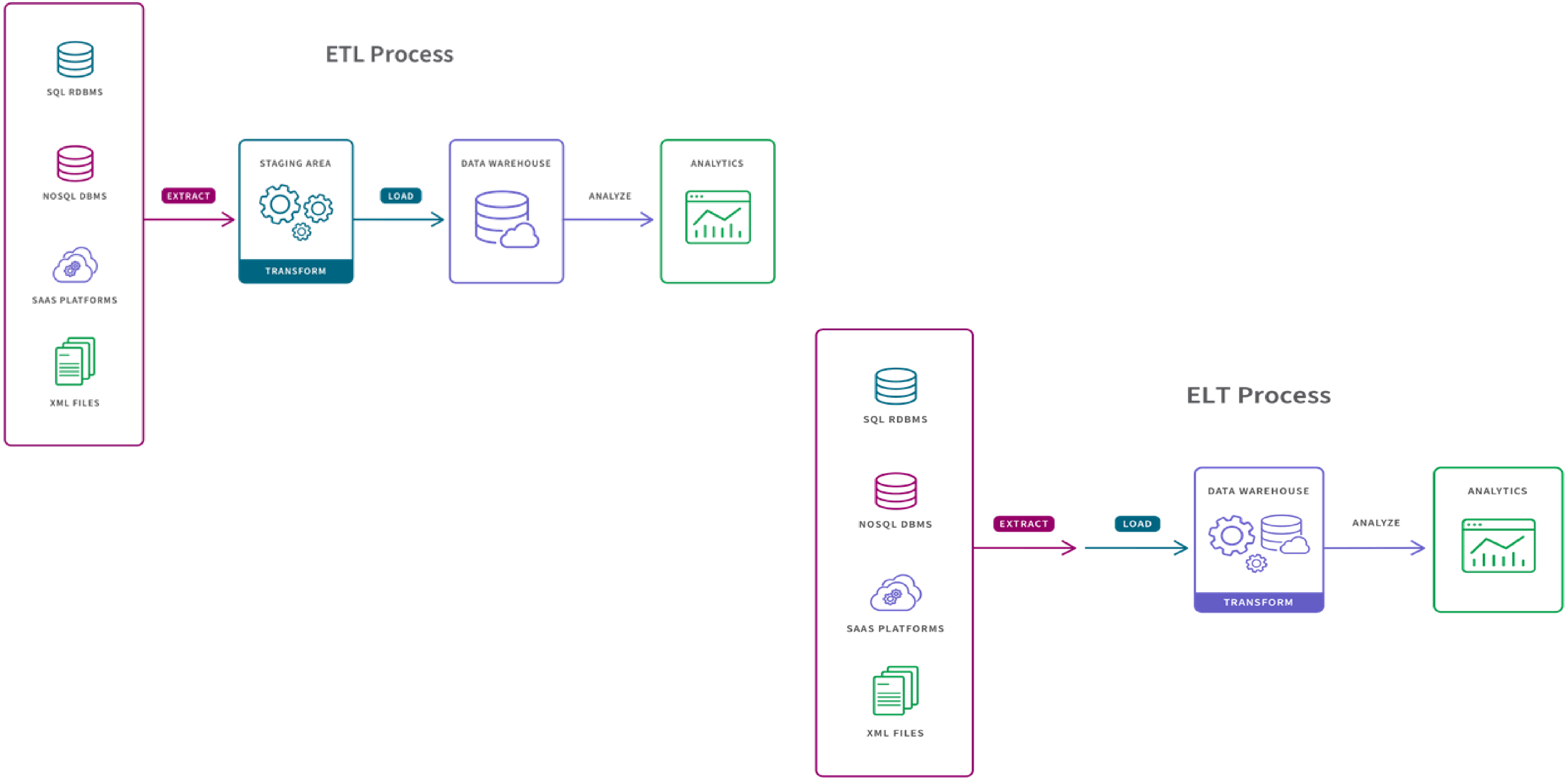


BREAK
15MINS

# Data Acquisition

Consists of data collection, data transmission, and data pre-processing.



# Data Storage: Database

A database is an organized collection of data, typically stored in a digital format, that is structured and managed to enable efficient storage, retrieval, and manipulation of information.

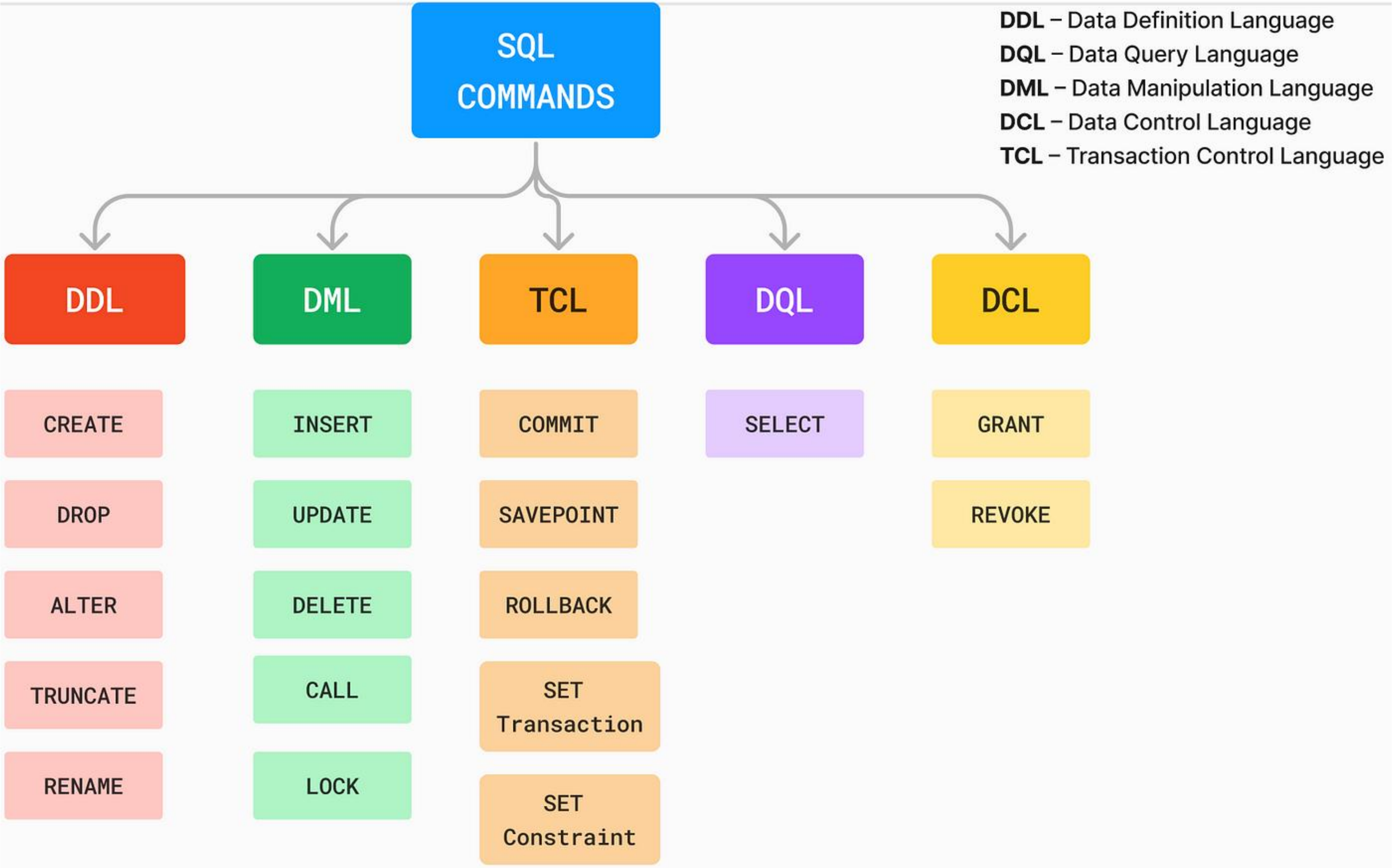

Databases are normally controlled using a database management system (DBMS).

In the database, data is organized into tables consisting of rows and columns.

Many databases also use Structured Query Language (SQL) for writing and querying data.


Components of a database

# Basic SQL Commands



# SQL Companies

Microsoft SQL Server

PostgreSQL

IBM DB2

MySQL

ORACLE

# Transactional vs. Analytical Database

## Transactional Database

- Supports management of an organization's data
- For everyday transactions

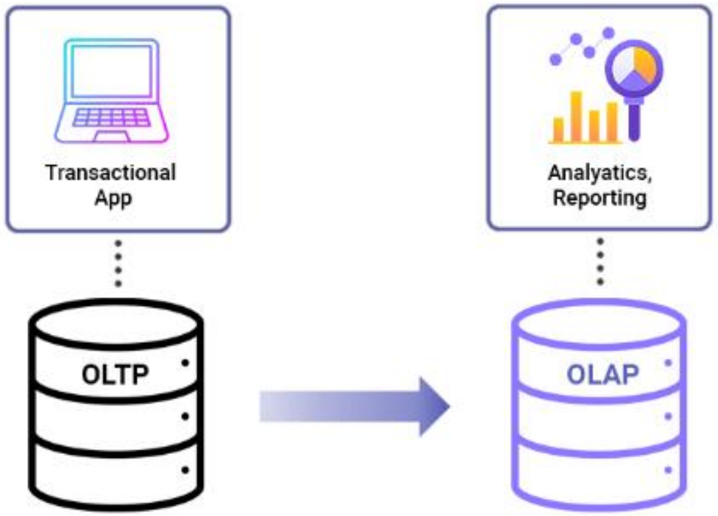
This is what is commonly thought of as "database management"

## Analytical Data Store

- Supports managerial decision-making
- For periodic analysis

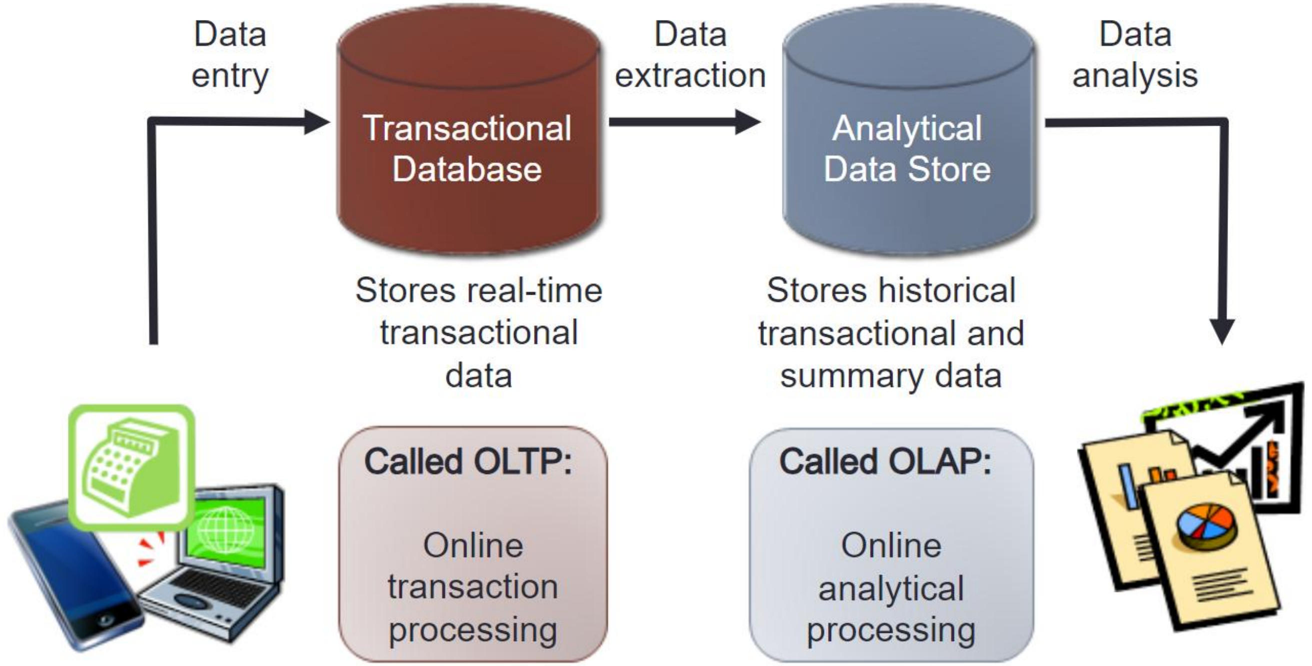
This is the foundation for business intelligence

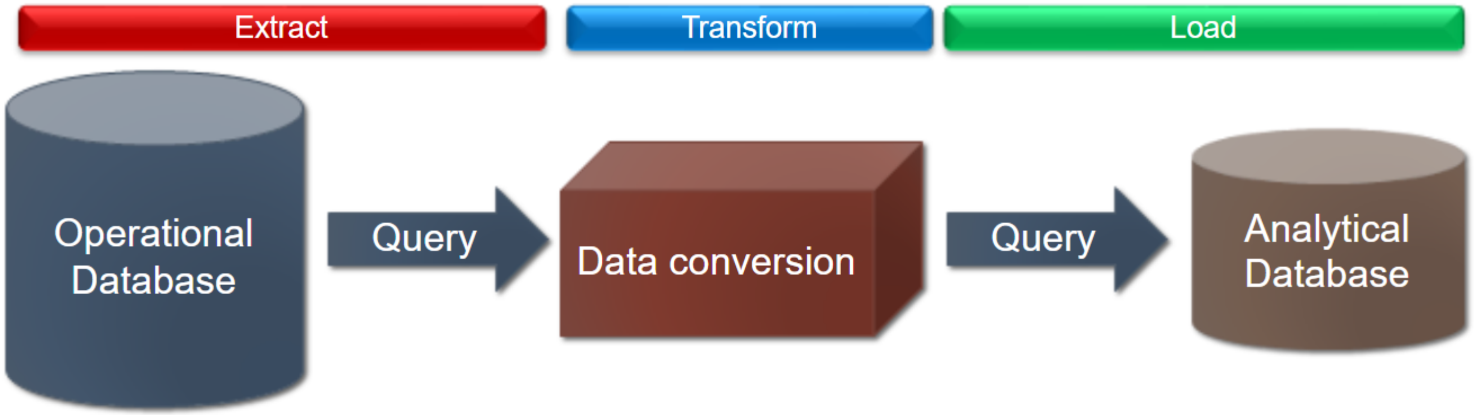
- High volume of transactions
- Fast processing
- Normalized data
- Many tables
- *"Who bought X?"*




- High volume of data
- Slow queries
- Denormalized data
- Fewer tables
- *"How many people bought X?"*

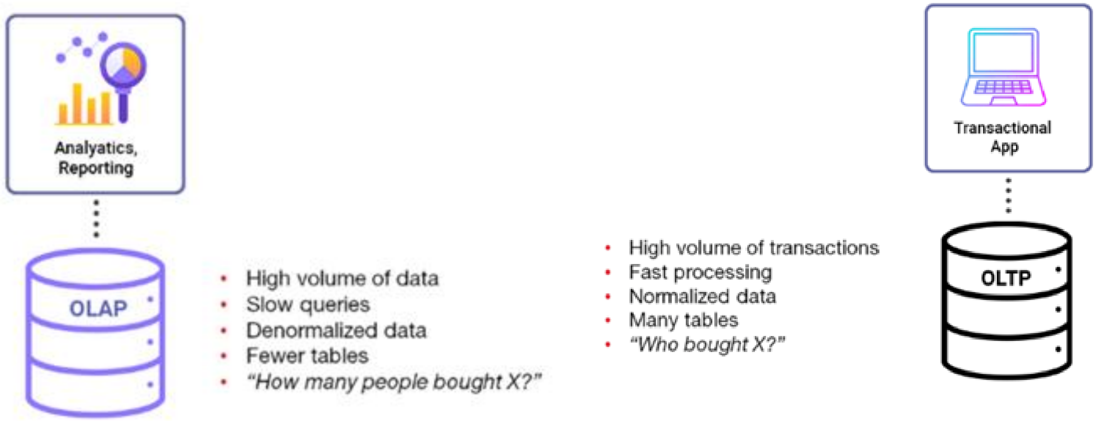
# Transactional vs. Analytical Database



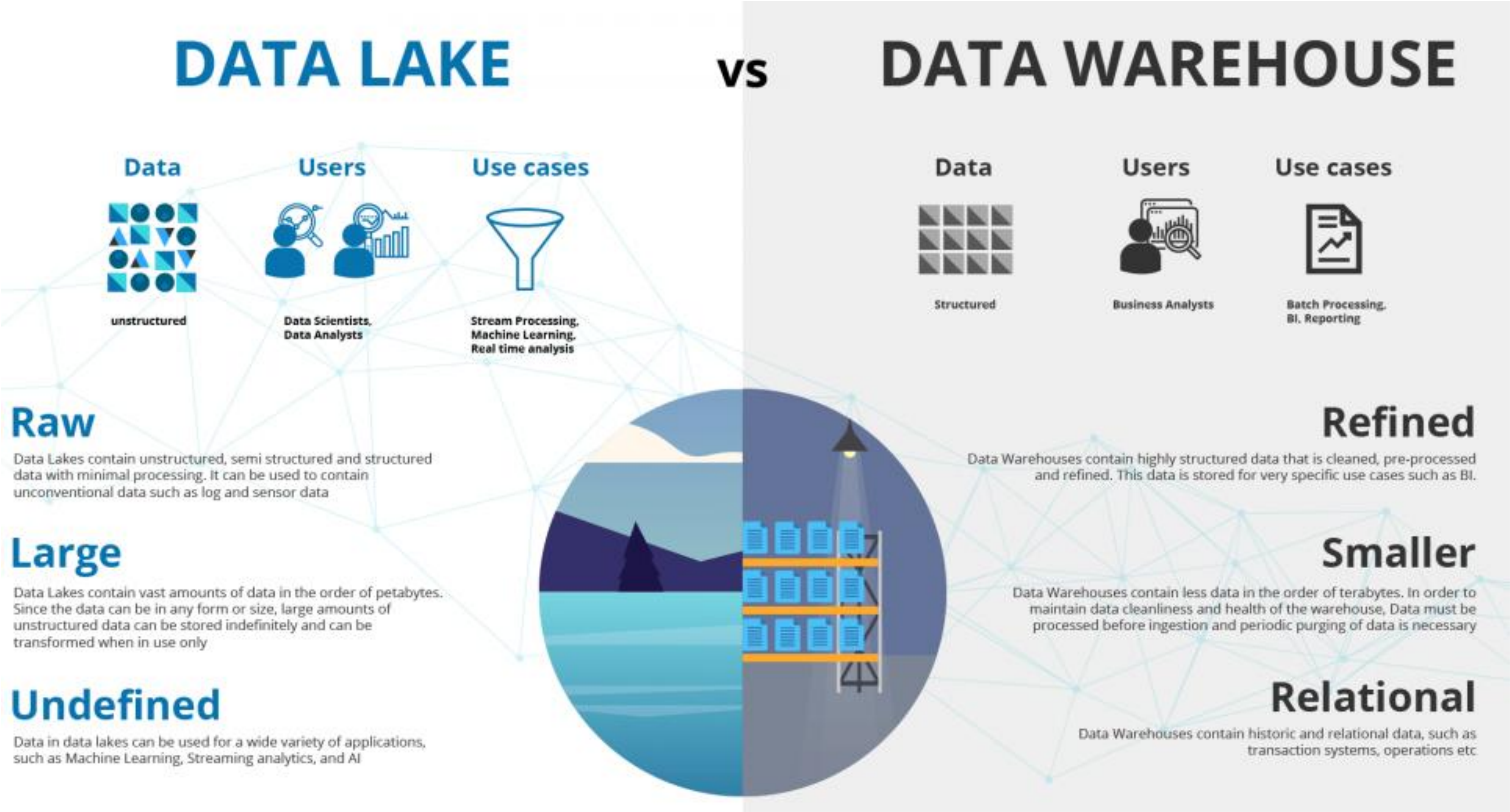
# Data Storage: Data Warehouse vs. Database

- A data warehouse is a large, central location where data is managed and stored for analytical processing.
- The data is accumulated from various sources and storage locations within an organization.
- For example, inventory numbers and customer information are likely managed by two different departments.
- However, both data types can be collected and presented on the same dashboard in a data warehouse.
- Then, data science professionals analyze the data for patterns and use their findings to help organizations make informed business decisions




| | Data warehouse | Database |
|---|---|---|
| Purpose | Analysis | Reporting |
| Database | OLAP (online analytical processing) | OLTP (online transactional processing) |
| Type of collection | Subject-oriented | Application-oriented |
| Query | Complex analytical queries | Simple transaction queries |

# Data Storage: Data Lake vs. Data Warehouse




Ref: https://www.qubole.com/data-lakes-vs-data-warehouses-the-co-existence-argument

# Data Storage: Data Lake vs. Data Warehouse

| Parameters | Data Lake | Data Warehouse |
|---|---|---|
| Data type | Raw (all types, no matter source of structure) | Processed (data stored according to metrics and attributes) |
| Data purpose | To be determined | Currently being used |
| Process | Extract Load Transform (ELT) | Extract Transform Load (ETL) |
| Schema position | After data storage, to offer agility and easy data capture | Before data storage, to offer security and high performance |
| Users | Data scientists, those who need in-depth analysis and tools (such as predictive modeling) to understand it | Business professionals, those who need it for operations |
| Accessibility | Accessible and easy to update | Complicated to make changes |
| History | Relatively new for big data | The concept has been around for decades |

Ref: https://www.guru99.com/data-lake-vs-data-warehouse.html

# Data Storage: Database vs. Data Warehouse vs. Data Lake


Ref: https://youtu.be/-bSkREem8dM?si=AqRddIFoihHWx7Sw/

# Challenges of Big Data


Ref: https://bleuwire.com/5-biggest-big-data-challenges/

# Challenges of RDBMS for Big Data

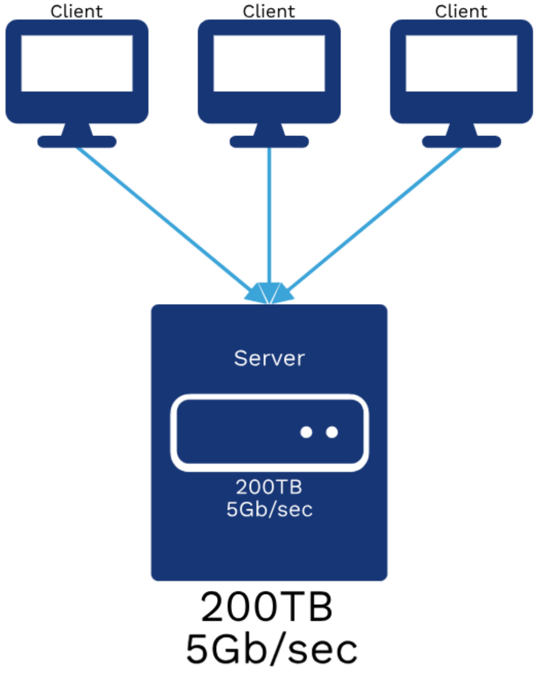
| Challenges of RDBMS for Big Data | Description |
|---|---|
| **Scalability** | Difficulty in scaling horizontally to handle large data volumes. |
| **Performance** | Slower query performance due to complex joins and indexing in large datasets. |
| **Data Variety** | Limited ability to handle diverse data types and formats (structured vs. unstructured). |
| **Data Velocity** | Inability to process real-time data streams efficiently. |
| **Cost** | High costs associated with licensing, maintenance, and hardware for scaling. |
| **Complexity** | Increased complexity in managing and optimizing RDBMS for large datasets. |
| **Schema Rigidity** | Fixed schema makes it difficult to adapt to changing data requirements. |
| **Data Distribution** | Challenges in distributing data across multiple nodes for load balancing. |

# NoSQL Databases

# BREAK
# 15MINS

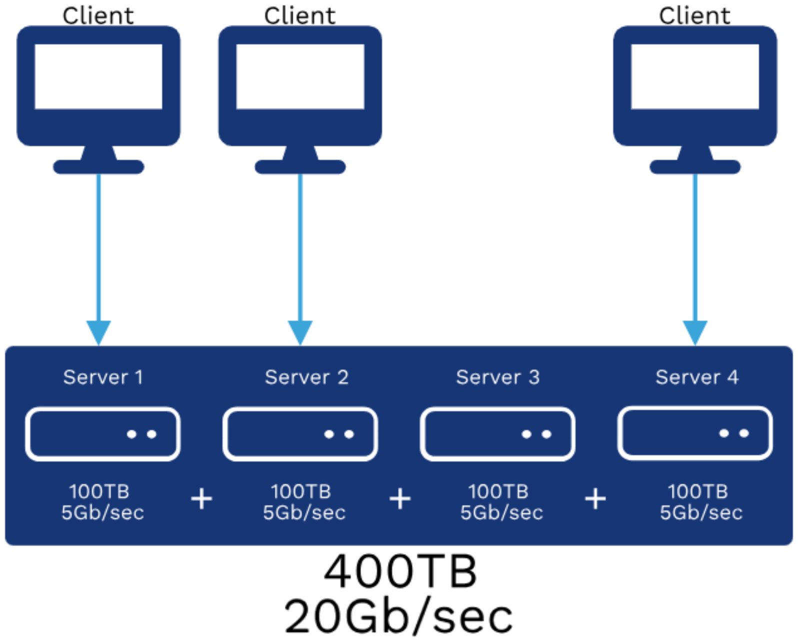
# How Can You Store Extremely Large Files?

## Centralized Database



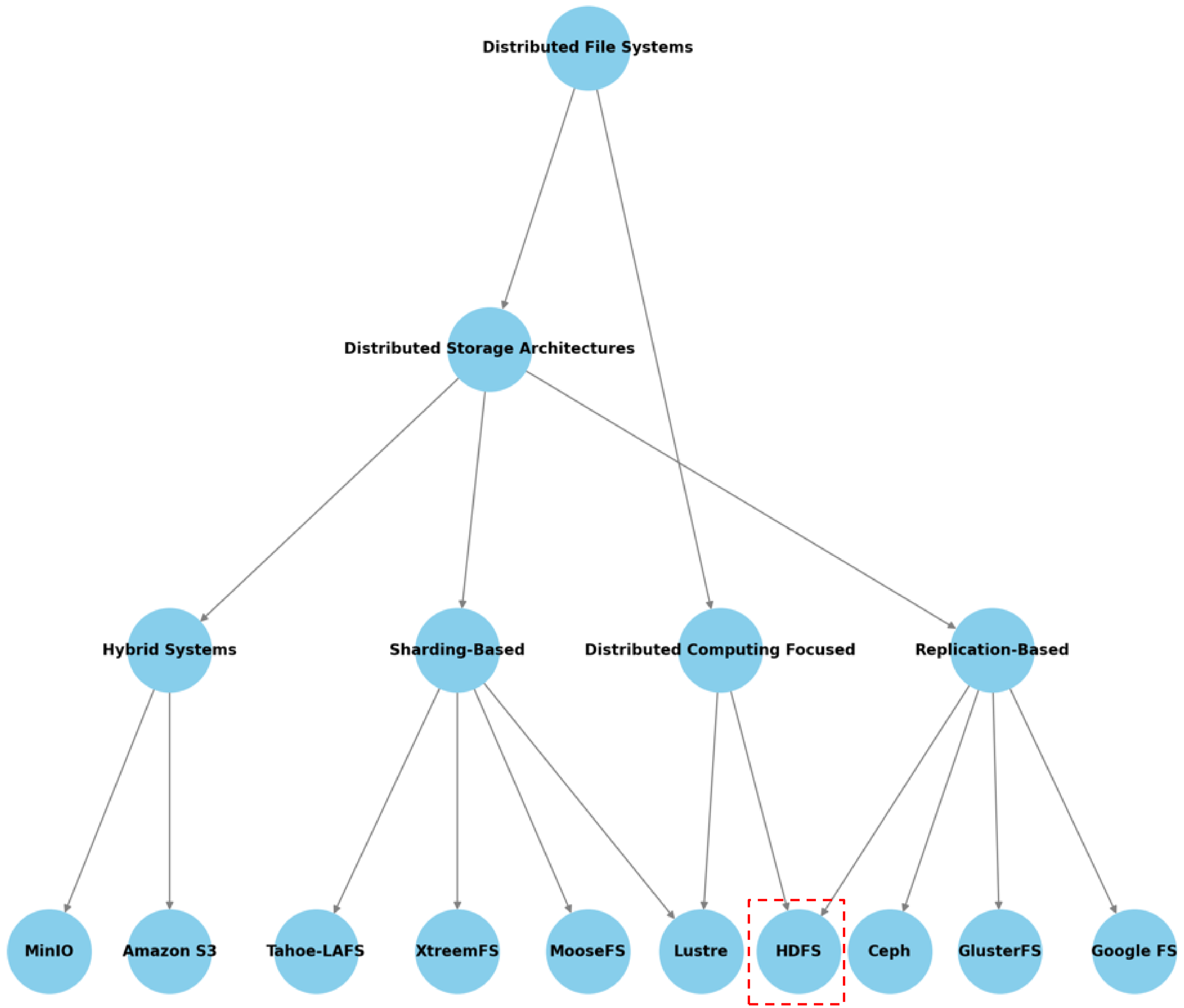

- A centralized database is stored and managed within a single location, with users accessing this data from wherever they are.
- This data is accessible by either the local area network (LAN) or wide area network (WAN).
- No copies of the data need to be made for users to find what they're looking for, as everything is stored in one place.

## Distributed Database



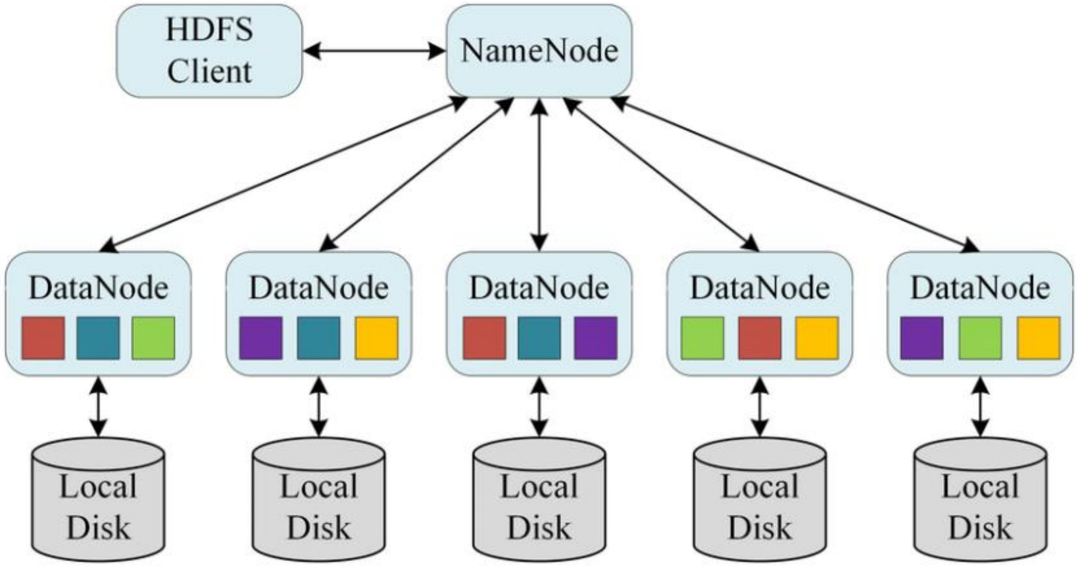

- With **distributed databases**, information is stored in multiple individual databases linked together by the network.
- These databases can all be stored in different physical locations and accessed from multiple networks.
- While this can keep information more secure, it's easy for data to be replicated and to create data redundancies.

# Heirarchy of DFS



# Hadoop Distributed File System (HDFS)

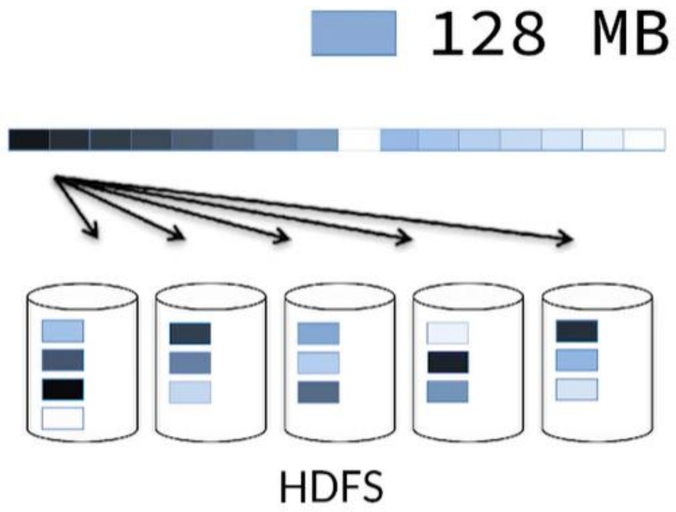
- HDFS is a "distributed file system designed to store and manage large amounts of data across multiple servers" in a Hadoop cluster.
- It is a critical component of the Hadoop ecosystem and is used to store and manage large data sets that are processed by Hadoop MapReduce or other Hadoop-based application



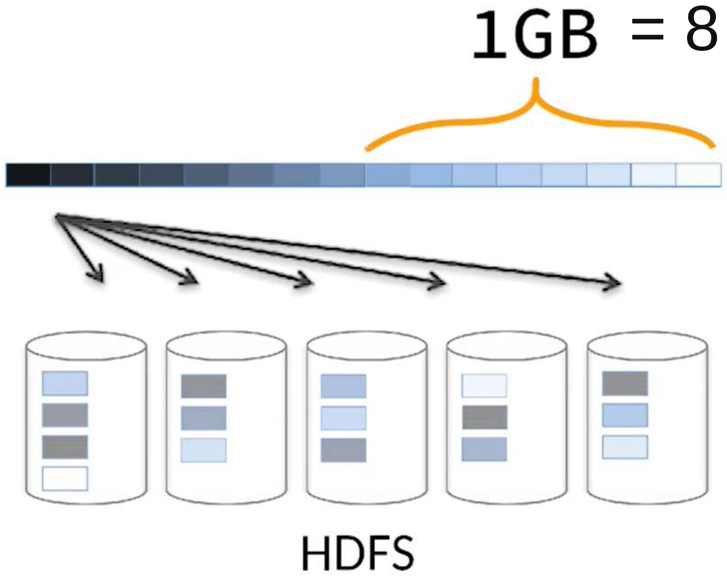

HDFS Features:

- HDFS is highly fault-tolerant and provides high throughput access to data.
- It is designed to run on low-cost hardware and can scale out to thousands of nodes in a single cluster.
- The data stored in HDFS is replicated across multiple nodes for reliability and high availability.

# HDFS




= 1 Block




= 8 Blocks


= 8000 Blocks

Each block is stored in 3 different discs

Yahoo! Hadoop Cluster

# Cost to store 1 PB of Big Data

- **Hard Disk Drives (HDD):**
  **US $0.03/GB * 1,000,000 GB * 4 = US $120,000**
- **Solid State Drives (SSD):**
  **US $0.20/GB * 1,000,000 GB * 4 = US $800,000**

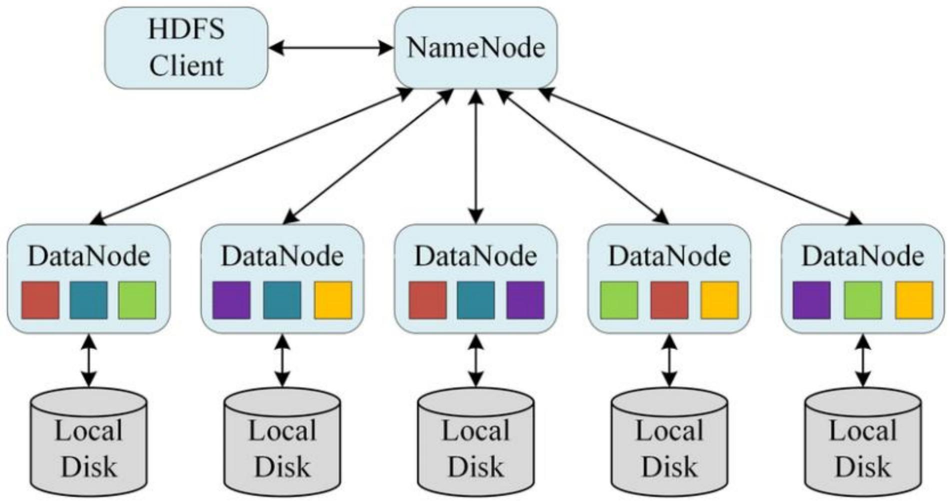
# HDFS Architecture

HDFS architecture consists of three main components:

**NameNode (Master Node):**
"The NameNode is the central node of HDFS", which manages the file system namespace, stores metadata information, and keeps track of the location of data blocks. It acts as a single point of failure, and its loss can result in the unavailability of the entire HDFS cluster.

**DataNode (Slave Node):**
"DataNodes are the worker nodes that store the actual data in HDFS". They are responsible for "storing", "retrieving", and "replicating data blocks" as directed by the NameNode. Each DataNode stores data locally and communicates with the Name Node to report its status, such as the amount of free space and the number of data blocks stored.

**Client:**
Clients are the applications or processes that access HDFS data. They communicate with the NameNode to locate the data they need and then directly communicate with the DataNodes to read or write data.

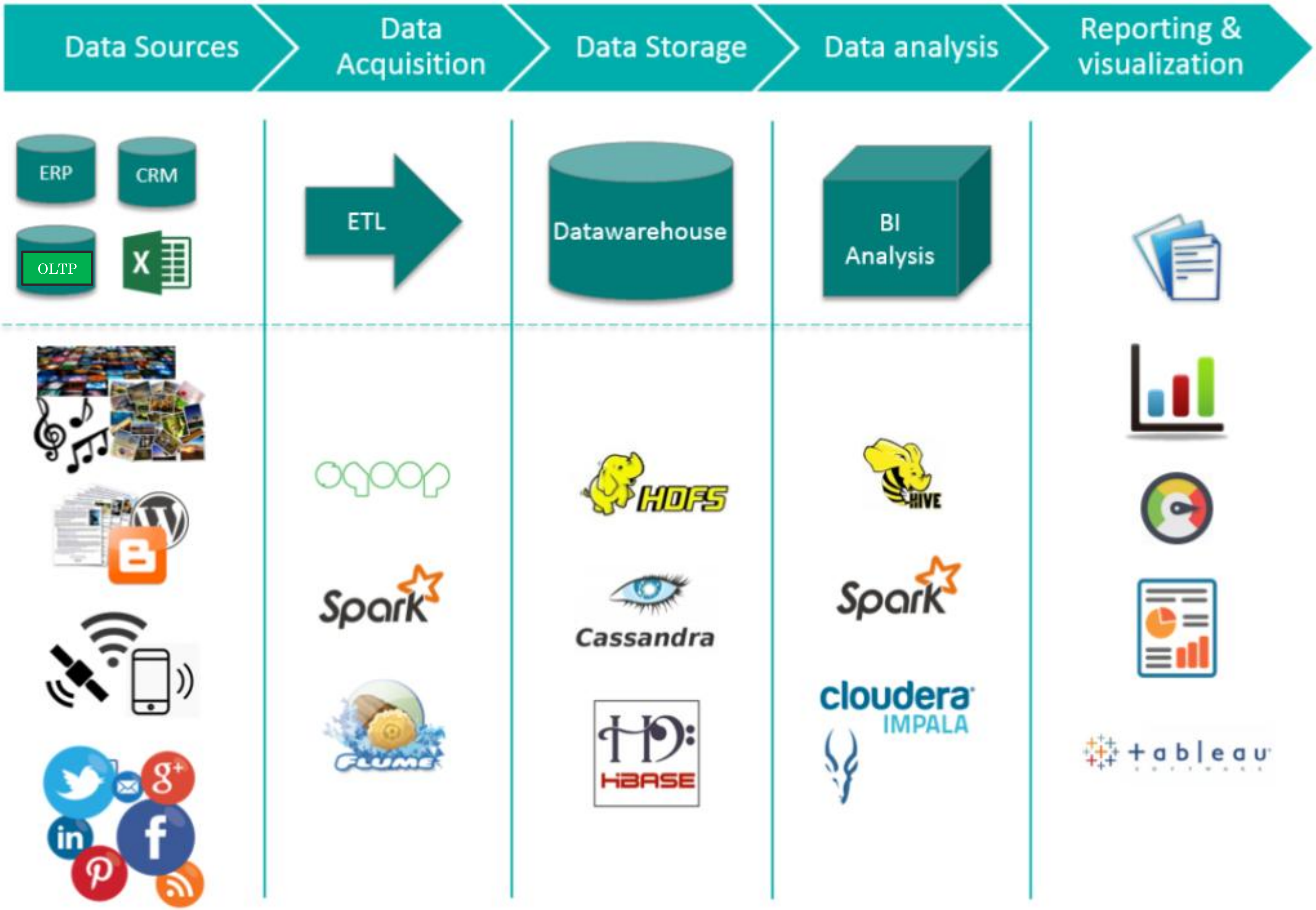


# Trivia Time

Which of the following is a major difference between storing data for a small database and for big data (hundreds of terabytes or more)?

- Storage for big data is distributed across a large number of disks
- Storage for big data should be done in the cloud to reduce the amount of network traffic
- Storage for big data has a limit of 1 exabyte, while storage for a small database is limited to tens of terabytes
- Storage for big data should be done on premises so you can use solid state disks

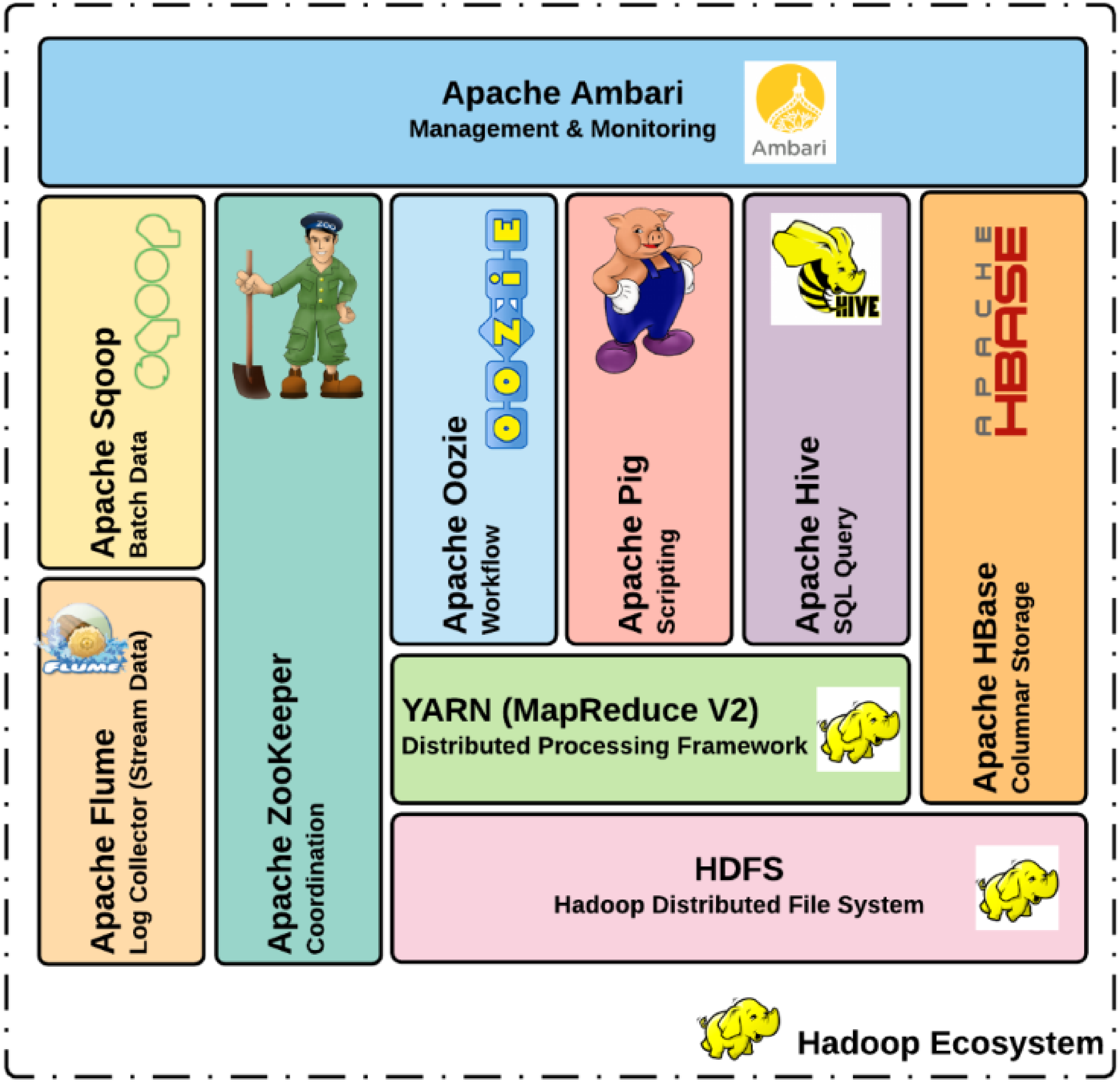
# Big Data Tools and Eco System




Ref: https://www.clearpeaks.com/big-data-ecosystem-spark-and-tableau/

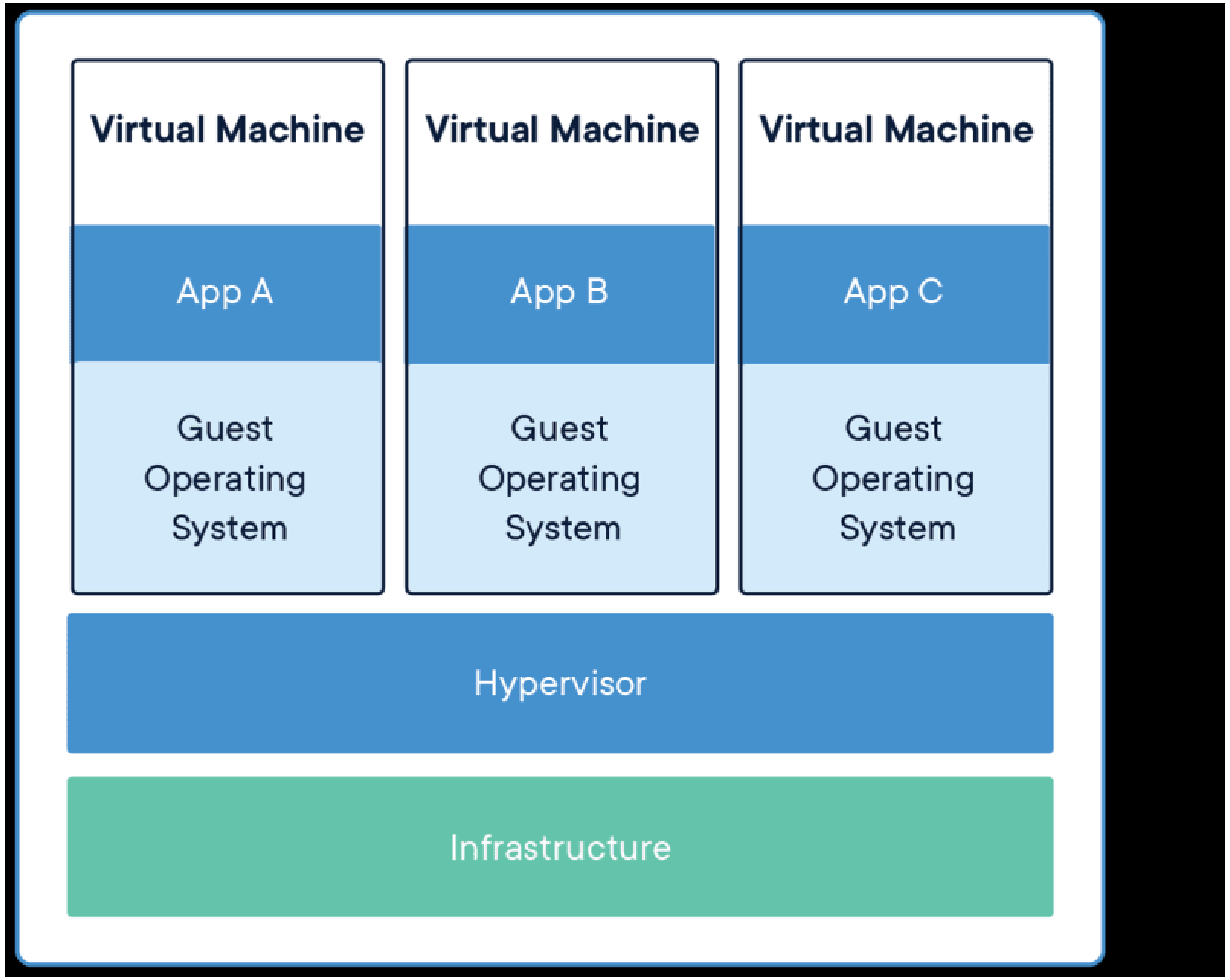
# Hadoop Ecosystem

"Hadoop is the framework and HDFS is the distributed filesystem used in this framework”



# What is a Virtual Machine?

A **Virtual Machine (VM)** is a software-based emulation of a physical computer. It runs an operating system (OS) and applications just like a real machine, but it operates in a sandboxed environment on a physical host system. Multiple VMs can run on a single physical machine using a **hypervisor**, which allocates resources like CPU, memory, and storage to each VM.



# Installing Virtual Machine

https://www.virtualbox.org/

Installing Instructions:

# Mentimeter Quiz

Mentimeter

Get Ready to Test Your Knowledge

# Thank You for Attention